

RANK Improvement WORKBOOK

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Electrical Engineering

Digital Electronics



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RANK Improvement WORKBOOK

Electrical Engineering

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Digital Electronics

IMPORTANT FORMULAE

1. Boolean Algebra, Logic Gates

EX-OR Gate

Boolean function of 2-input EXOR operation

$$Y = A \oplus B = \bar{A}B + A\bar{B}$$

$$A \oplus A = 0, \quad A \oplus 0 = A$$

$$A \oplus \bar{A} = 1, \quad A \oplus 1 = \bar{A}$$

EX-NOR Gate

Boolean function of 2-input EXNOR operation

$$Y = A \odot B = \overline{A \oplus B} = \overline{(\bar{A}B + A\bar{B})} = AB + \bar{A}\bar{B}$$

$$A \odot A = 1, \quad A \odot 1 = A$$

$$A \odot \bar{A} = 0, \quad A \odot 0 = \bar{A}$$

$$\bar{A} \oplus B = A \odot B \text{ and } A \oplus \bar{B} = A \odot B$$

$$\bar{A} \odot B = A \oplus B \text{ and } A \odot \bar{B} = A \oplus B$$

Boolean Algebraic Laws

Commutative Law:

$$A + B = B + A \quad \text{and} \quad A \cdot B = B \cdot A$$

Associative Law:

$$A + (B + C) = (A + B) + C = A + B + C$$

$$A \cdot (B \cdot C) = (A \cdot B) \cdot C = A \cdot B \cdot C$$

Distributive Law:

$$A(B + C) = AB + AC$$

$$(A + B)(C + D) = AC + AD + BC + BD$$

Boolean Algebraic Theorems

AND-Operation Theorem:

$$A \cdot A = A \quad A \cdot 0 = 0$$

$$A \cdot 1 = A \quad A \cdot \bar{A} = 0$$

Involution Theorem:

$$(A')' = \bar{\bar{A}} = A$$

OR-Operation Theorem:

$$A + A = A \quad A + 0 = A$$

$$A + 1 = 1 \quad A + \bar{A} = 1$$

De Morgan's Theorem:

$$\overline{(A_1 \cdot A_2 \cdot A_3 \cdot \dots \cdot A_n)} = \bar{A}_1 + \bar{A}_2 + \bar{A}_3 + \dots + \bar{A}_n$$

$$\overline{(A_1 + A_2 + A_3 + \dots + A_n)} = \bar{A}_1 \cdot \bar{A}_2 \cdot \bar{A}_3 \cdot \dots \cdot \bar{A}_n$$

Transposition Theorem:

$$(A + B)(A + C) = A + BC$$

Distribution Theorem:

$$A + BC = (A + B)(A + C)$$

Boolean Algebraic Theorems

Theorem No.	Theorem
1.	$(A + B) \cdot (A + \bar{B}) = A$
2.	$AB + \bar{A}C = (A + C)(\bar{A} + B)$
3.	$(A + B)(\bar{A} + C) = AC + \bar{A}B$
4.	$AB + \bar{A}C + BC = AB + \bar{A}C$
5.	$(A + B)(\bar{A} + C)(B + C) = (A + B)(\bar{A} + C)$
6.	$\overline{A \cdot B \cdot C \cdot \dots} = \bar{A} + \bar{B} + \bar{C} + \dots$
7.	$\overline{A + B + C + \dots} = \bar{A} \cdot \bar{B} \cdot \bar{C} \cdot \dots$

For N -variables, maximum possible Self-Dual Function = $(2)^{2^{n-1}} = 2^{(2^n/2)}$.

With N -variables, maximum possible distinct logic functions = 2^{2^n} .

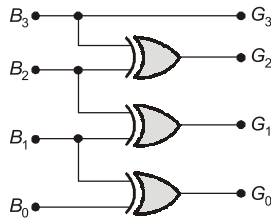
2. Combinational Circuits

Binary-to Gray Code Converter

Equivalent logical gate diagram:

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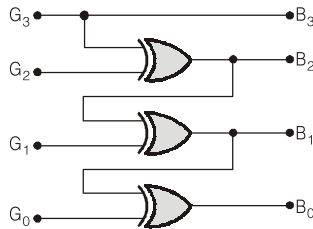
$$\begin{aligned} G_3 &= B_3 \\ G_2 &= B_3 \oplus B_2 \\ G_1 &= B_2 \oplus B_1 \\ G_0 &= B_1 \oplus B_0 \end{aligned}$$



Gray to Binary Converter

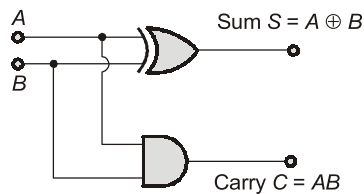
Equivalent Logical Gate Diagram:

$$\begin{aligned} B_3 &= G_3 \\ B_2 &= B_3 \oplus G_2 \\ B_1 &= B_2 \oplus G_1 \\ B_0 &= B_1 \oplus G_0 \end{aligned}$$



Combinational Circuits

Logic circuit:



Logical expression:

$$\text{Sum } S = \bar{A}B + A\bar{B} = A \oplus B$$

$$\text{Carry } C = AB$$

Remember:

- Total number of NAND-gates required to implement half adder = 5
- Total number of NOR-gates required to implement half adder = 5
- To implement the half adder circuit by minimum number of logic gates (if we have all gates except EXOR and EXNOR) is "3".
- Total number of MUX required to implement half adder = 3.

Full Adder (F.A.):

$$\text{Sum, } S = A \oplus B \oplus C$$

$$\text{Carry, } C = AB + BC + CA = AB + C(A \oplus B)$$

Remember:

- A F.A. can be implemented by two H.A. and one OR-gate
- Total number of NAND-gate/NOR-gate required to implement a F.A is equals to "9".
- Total number of MUX required to implement full adder = 7.

Half Subtractor (H.S)

$$\text{Difference, } D = \bar{A}B + A\bar{B} = A \oplus B$$

$$\text{Borrow, } B_o = \bar{A}B$$

Remember:

- Total number of NAND/NOR gates required to implement the H.S is equal to "5".
- Total number of MUX required to implement half subtractor = 3.

Full Subtractor (F.S):

$$\text{Difference, } D = A \oplus B \oplus C$$

$$\text{Borrow, } B_o = \bar{A}B + \bar{A}C + BC = \bar{A}B + (\bar{A} \oplus B) \cdot C$$

Remember:

- A full subtractor can be implemented with two half subtractor and one OR Gate.
- Number of NAND/NOR gates required to implement the full subtractor is equal to '9'
- In parallel adder n full adder or {(n - 1) F.A. and 1 H.A.} or {(2n - 1) H.A. and (n - 1) OR Gate} are required to add two n-bit numbers.
- Total number of MUX required to implement full subtractor = 7.

Look Ahead Carry Adder

Generator signal,

$$G_i = A_i B_i$$

Propagation signal,

$$P_i = A_i \oplus B_i$$

The carry signal generated is

$$C_{i+1} = P_i C_i + G_i$$

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Number of AND gates required in carry circuit for n -bit adder is

$$N = \frac{n(n+1)}{2}$$

Number of OR gates required in carry circuit of n -bit adder is $N = n$.

Higher order MUX using Lower order MUX:

Given MUX	To be Implemented MUX	Required number of MUX
2 : 1	4 : 1	3
4 : 1	16 : 1	4 + 1 = 5
4 : 1	64 : 1	16 + 4 + 1 = 21
8 : 1	64 : 1	8 + 1 = 9
8 : 1	256 : 1	32 + 4 + 1 = 37

Remember:

- To implement $2^n : 1$ MUX by using 2 : 1 MUX, the total number of 2 : 1 MUX required is $(2^n - 1)$.
- MUX is a Universal Logic gate.
- Using one $2^n : 1$ MUX

Higher Order DEMUX Using Lower Order DEMUX:

Given DEMUX	To be Implemented MUX	Required number of MUX
1 : 2	1 : 4	1 + 2 = 3
1 : 2	1 : 8	1 + 2 + 4 = 7
1 : 2	1 : 16	1 + 2 + 4 + 8 = 15
1 : 2	1 : 64	1 + 2 + 4 + 8 + 16 + 32 = 63
1 : 4	1 : 16	1 + 4 = 5

3. Sequential Circuits

Conversion of Flip-Flops

SR FF to other FFs	JK FF to other FFs	D-FF to other FFs	T-FF to other FFs
$S = J\bar{Q}_n$ & $R = KQ_n$	$J = S$ & $K = R$	$D = S + \bar{R}Q_n$	$T = S\bar{Q}_n + RQ_n$
$S = D$ & $R = \bar{D}$	$J = D$ & $K = \bar{D}$	$D = J\bar{Q}_n + \bar{K}Q_n$	$T = J\bar{Q}_n + KQ_n$
$S = T\bar{Q}_n$ & $R = TQ_n$	$J = K = T$	$D = T \oplus Q_n$	$T = D \oplus Q_n$

Excitation Table of Flip-Flops

Q_n	Q_{n+1}	S	R	J	K	D	T
0	0	0	X	0	X	0	0
0	1	1	0	1	X	1	1
1	0	0	1	X	1	0	1
1	1	X	0	X	0	1	0

Time delay for registers

A SISO SRs may be used to introduce time delay " Δt " in digital signals

$$\Delta t = N \times T = N \times \frac{1}{f_c}$$

Remember for counters:

- If N = total number of states and ' n ' = number of FFs then

$$N \leq 2^n$$

$$n \geq 3.32 \log_{10} N \quad \text{or} \quad n \geq \log_2 N$$

- If $N = 2^n$, then we get BINARY COUNTER.
- If $N < 2^n$, then we get NON-BINARY COUNTER.

Binary Ripple Counter

- With n -FFs the maximum count that can be counted by this counter is $(2^n - 1)$.
- If input frequency is ' f ' then output frequency

$$\text{is } \frac{f}{2^n}.$$

- It is also known as $2^n : 1$ scalar divider.

For proper operation of the ripple counter:

$$T_{CLK} \geq nt_{pd(FF)}$$

$$f_{CLK} \leq \frac{1}{nt_{pd(FF)}}$$

Maximum CLK frequency:

$$f_{CLK,m} = \frac{1}{nt_{pd(FF)}}$$

For determination of Up/Down Counter:

Triggering with	CLK connection in	Access as
(-ve) edge	Q	UP Counter
(-ve) edge	$\bar{Q}$	Down Counter
(+ve) edge	Q	Down Counter
(+ve) edge	$\bar{Q}$	UP Counter

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Synchronous-Series Carry Counter

Clock frequency:

$$f_{CLK} \leq \frac{1}{t_{pd(FF)} + (n-2)t_{pd(AND\ gate)}}$$

Synchronous-Parallel Carry Counter

Clock frequency:

$$f_{CLK} = \frac{1}{t_{pd(FF)} + t_{pd(AND\ gate)}}$$

4. A/D and D/A Converters**Digital to Analog Converter (DAC)**

Resolution:

$$\text{Resolution} = \frac{V_{FS}}{2^n - 1}$$

Analog Output Voltage:

$$V_0 = K [2^{N-1} b_{N-1} + 2^{N-2} b_{N-2} + \dots + 2^2 b_2 + 2^1 b_1 + b_0]$$

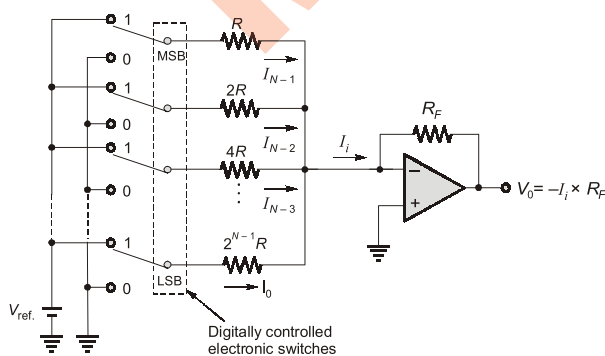
Full scale output (V_{FS}) voltage is maximum output of DAC $V_{FS} = \text{Resolution} \times \text{Maximum decimal}$

$$= \frac{V_{FS}}{2^n - 1} \times (2^n - 1)$$

% Resolution:

$$\% \text{ Resolution} = \frac{\text{Resolution}}{V_{FS}} \times 100$$

$$\% \text{ Resolution} = \frac{1}{2^n - 1} \times 100$$

Weighted-Resistor DAC

Output voltage:

$$V_0 = \frac{R_F}{2^{N-1} \cdot R} (2^{N-1} V_{N-1} + 2^{N-2} V_{N-2} + \dots + 2^1 V_1 + 2^0 V_2)$$

Input current to OP-AMP:

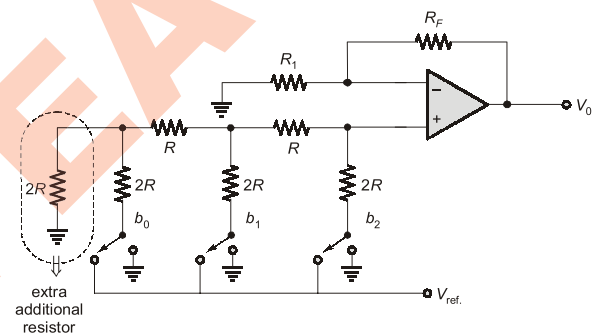
$$I_i = \frac{V_{ref.}}{2^{N-1} \cdot R} \sum_{i=0}^{N-1} 2^i b_i$$

Maximum output current:

$$I_{max} = \frac{V_{ref.}}{2^{N-1} \cdot R} (2^N - 1)$$

Resistance:

$$\text{LSB Resistance} = (2^{N-1}) \times \text{MSB Resistance}$$

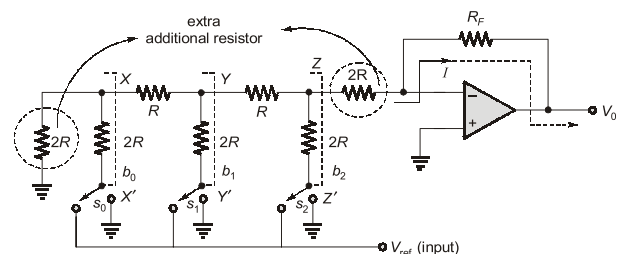
R-2R Ladder DAC by Using Non-inverting OP-AMP

Gain of OP-AMP

$$\frac{V_0}{V_{ref.}} = \left(1 + \frac{R_F}{R_1} \right)$$

Output analog voltage

$$V_0 = \frac{V_{ref.}}{2^N} \left[\sum_{i=0}^{N-1} 2^i b_i \right] \times \left[1 + \frac{R_F}{R_1} \right]$$

R-2R Ladder DAC by Using Inverting OP-AMP

Output voltage:

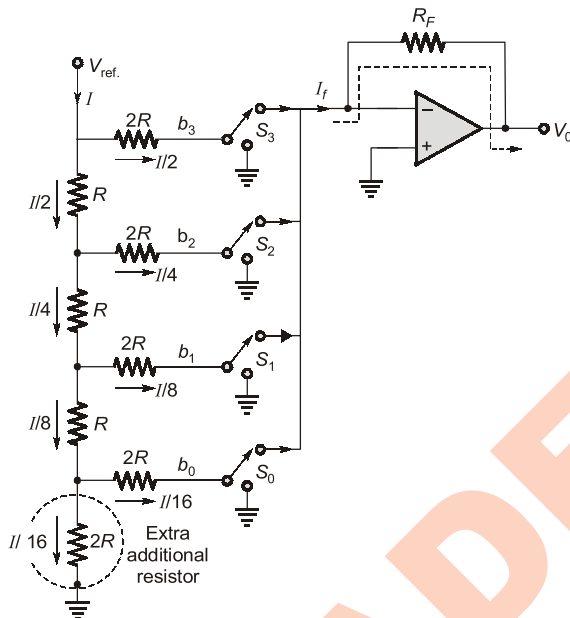
For N-bit DAC

$$V_0 = \frac{V_{\text{ref.}}}{2^N} \times \left[\sum_{i=0}^{N-1} 2^i b_i \right] \times \left[\frac{-R_F}{3R} \right]$$

Output current:

$$I = \frac{V_{\text{ref.}}}{2^N} \times \left[\sum_{i=0}^{N-1} 2^i b_i \right] \times \left[\frac{1}{3R} \right]$$

Inverted Ladder (R-2R) type DAC



Output voltage:

For N-bit DAC

$$V_0 = \frac{V_{\text{ref.}}}{2^N} \times \left[\sum_{i=0}^{N-1} 2^i b_i \right] \times \left(\frac{-R_F}{R} \right)$$

Forward current:

$$I_f = \frac{V_{\text{ref.}}}{2^N} \times \left[\sum_{i=0}^{N-1} 2^i b_i \right] \times \frac{1}{R}$$

Voltage Range:

$$\text{Voltage range} = V_{\text{max}} - V_{\text{min}}$$

Resolution:

$$\text{Resolution} = \frac{V_{\text{range}}}{2^n - 1}$$

$$\% \text{ Resolution} = \frac{1}{2^n - 1} \times 100$$

Dynamic range

$$\text{Dynamic range} = (1.8 + 6n) \text{ dB}$$

Counter type ADC:

- Number of clock pulse required for n-bit conversion = $2^n - 1$
- Maximum conversion time = $(2^N - 1) T_{\text{clk}}$

SAR type ADC:

- Number of clock pulse required for n-bit conversion = n .
- Maximum conversion time = $n * T_{\text{clk}}$
- In SAR ADC, conversion time is independent of input analog voltage.

Flash type ADC:

For n bit conversion flash type ADC requires

- $2^n - 1$ comparators
- 2^n resistors

Dual slope integrating type ADC:

- Total number of clock pulse required = $2^n + N \approx 2^{n+1}$.
- It is most accurate ADC therefore mostly used in digital voltmeter.
- It is slowest ADC.

Remember:

	No. of clocks	Time
Counter type	2^{n-1}	$(2^{n-1}) T_{\text{clk}}$
SAR	n	$n T_{\text{clk}}$
Counter type	1	T_{clk}
Counter type	2^{n+1}	$2^{n+1} T_{\text{clk}}$

5. Logic Families

Propagation Time Delay (t_{pd})

Delay time:

$$t_{pd} = \frac{t_{PHL} + t_{PLH}}{2} \text{ ns}$$

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Remember:

- The delay times are measured between the 50% points on the input and output transitions.
- In BJT, $t_{PLH} > t_{PHL}$ due to reverse recovery time.
- In FET, $t_{PLH} < t_{PHL}$ due to large capacitance formed.

Power Dissipation (P_D)

Collector current:

$$I_{CC} = \frac{I_{CCH} + I_{CCL}}{2}$$

Power dissipation:

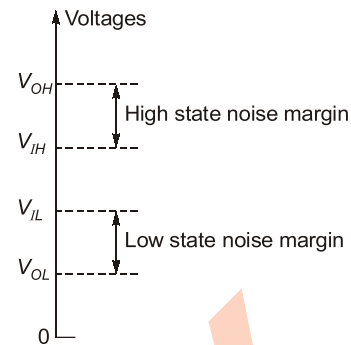
$$P_{D(avg)} = I_{CC} \times V_{CC} \text{ mW}$$

Figure of Merit (FOM)

$$\text{FOM} = t_{pd} \times P_{D(avg)} \text{ pico Joules}$$

Fan-in and Fan-out

$$\text{Fan out} = \left(\frac{I_{OH}}{I_{IH}} \right) \text{ or } \left(\frac{I_{OL}}{I_{IL}} \right)$$

Noise-Margin

$$V_{OH} > V_{IH} > V_{IL} > V_{OL}$$

$$\text{High state N.M.} = V_{NH} = V_{OH(\min)} - V_{IH(\min)}$$

$$\text{Low state N.M.} = V_{NL} = V_{IL(\max)} - V_{OL(\max)}$$

$$\text{Noise margins} = (V_{NH}, V_{NL})_{\min}$$

Power Dissipation

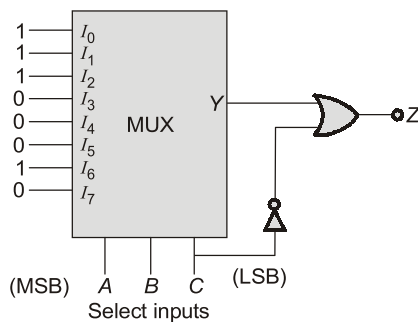
$$P_d(\text{dynamic}) = C_f V_{DD}^2$$

■■■■

Boolean Algebra, Logic Gates

LEVEL 1 Questions

- Q.1** A combinational circuit using a 8-to-1 multiplexer is shown in the figure. The minimized expression for the output (Z) is



- (a) $C(\bar{A} + \bar{B})$ (b) $C(A + B)$
 (c) $\bar{C} + \bar{A}\bar{B}$ (d) $\bar{C} + AB$
- Q.2** A boolean function is represented as $f(X, Y, Z) = XY + XZ$. The POS representation of the function in K-map can be best described by (Assuming no don't care condition is present)

(a)

YZ \ X	00	01	11	10
0				
1		0	0	0

(b)

YZ \ X	00	01	11	10
0		0	0	0
1			0	

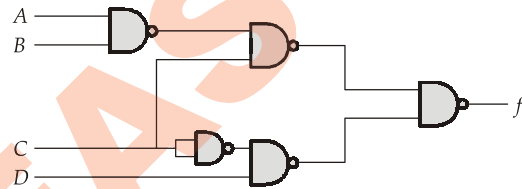
(c)

YZ \ X	00	01	11	10
0	0	0	0	0
1	0			

(d)

YZ \ X	00	01	11	10
0			0	
1	0	0	0	0

- Q.3** Consider the circuit shown in the figure below:



The circuit consists of NAND gates only. Then which one of the following options represents the output of the logic circuit?

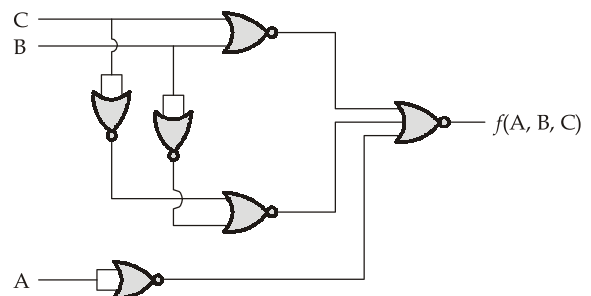
- (a) $f = (\bar{A} + \bar{B})C + \bar{C}D$
 (b) $f = (\bar{A} + \bar{B})\bar{C} + \bar{C}D$
 (c) $f = \overline{ABC} + \bar{D}C$
 (d) $f = (\bar{A} + \bar{B})\bar{C} + \bar{D}$

- Q.4** A Boolean function is given as,

$$f(A, B, C, D) = B\bar{D} + \bar{B}CD + ABC + AB\bar{C}D + \bar{B}\bar{D}$$

Then the minimum number of two input NAND gates required to implement this function is _____.

- Q.5** A function $f(A, B, C)$ is produced by using NOR-gates as shown in the figure below:



The function 'f' can be represented as

- (a) $f = \sum m(0, 1, 3, 2, 7)$
- (b) $f = \sum m(0, 1, 3)$
- (c) $f = \sum m(5, 6)$
- (d) $f = \sum m(1, 5, 7)$

Q.6 Let $f(A, B) = A' + B$, then which of the given options is the simplified form of the function $f(f(x + y, y), z)$?

- (a) $xy' + z$ (b) $x' + z$
- (c) $x + z$ (d) $x + z'$

Q.7 The expression $f = \overline{AB + DC + AB + A}$ when reduces, gives

- (a) 0 (b) 1
- (c) $\overline{A + DC}$ (d) $A + DC$

Q.8 Given that $306_{10} = 615$ in some number system. The base of the such system will be _____.

Q.9 Given $(6y3)_x + (54)_x = 487$, what can be the possible value of base x and y respectively

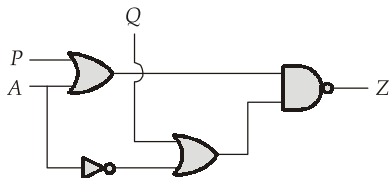
- (a) 10, 2 (b) 8, 7
- (c) 11, 3 (d) None of these

Q.10 The Boolean function $Y(A, B, C) = \overline{AB} + A \oplus C$.

The SOP and POS from will be

- (a) $Y = \sum(1, 2, 3, 6, 7)$ and $Y = \prod(0, 4, 5)$
- (b) $Y = \sum(1, 3, 4, 5, 6)$ and $Y = \prod(0, 2, 7)$
- (c) $Y = \sum(0, 2, 5, 6)$ and $Y = \prod(1, 3, 4, 7)$
- (d) None of the above

Q.11 The circuit shown below is used to implement the function $Z = f(A, B) = \overline{A} + B$. The values of P and Q are



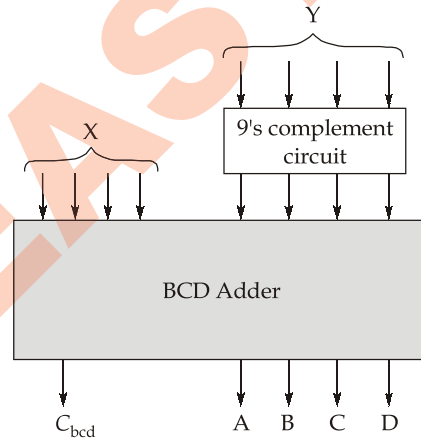
- (a) $P = A, Q = B$ (b) $P = B, Q = \overline{A}$
- (c) $P = \overline{B}, Q = 0$ (d) $P = 0, Q = \overline{B}$

Q.12 Consider the function $F(x, y, z) = xy + x'z + yz$. All the variables are available in complemented and non-complemented form. The minimum number of gates required to implement the above function is _____. (All gates are available only with 2-inputs)

Q.13 Given $F(A, B, C, D) = \sum m(0, 2, 3, 5, 7, 8, 10, 12, 13, 14, 15)$.

The number of prime implicants is _____.

Q.14 In the given below circuit input $X = 1001$ and $Y = 0011$ the output ABCD ($8 - 4 - 2 - 1$) and C_{bcd} of the circuit respectively are:



- (a) 1111, 0 (b) 1011, 0
- (c) 0100, 1 (d) 0101, 1

Q.15 The 6's complement of the number $(73)_8$, when it is expressed with the base of 6

- (a) 155 (b) 312
- (c) 135 (d) 421

Q.16 Let $x_1 \oplus x_2 \oplus x_3 = 1$, where x_1, x_2, x_3 are Boolean variables. Which one of the following always true?

- (a) $x_1 = x_2 \cdot x_3$ (b) $x_1 x_2 \cdot x_3 = 0$
- (c) $x_1 + x_2 + x_3 \neq 1$ (d) $x_1 \neq x_2 \oplus x_3$

Q.17 Consider the following Boolean expression for F.

$$F(A, B, C, D) = \overline{C}D + \overline{A}\overline{C}\overline{D} + AB\overline{C}\overline{D}$$

The minimal sum-of-products for the function $F(A, B, C, D)$ is

- (a) $\overline{A} + \overline{B} + \overline{C} + \overline{D}$ (b) $\overline{A}B + \overline{B}C + \overline{C}D$
- (c) $\overline{A}\overline{C} + \overline{C}D + B\overline{C}$ (d) $\overline{C}D + \overline{B} + \overline{D}$

Q.18 Which of the following logic gate is used as equality and inequality checking respectively?
(a) NAND and NOR (b) EX-NOR and EX-OR
(c) EX-NOR and OR (d) EX-OR and EX-NOR

Q.19 The representation of the value of a 16-bit unsigned integer X in hexadecimal number system is A72E. The representation of the value of X in octal number system is
(a) 12346 (b) 123456
(c) 125756 (d) 10634

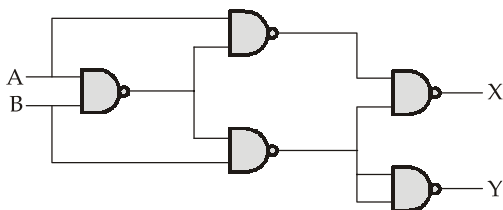
Q.20 If $(504)_X$ in base-X is equal to $(2320)_4$. Then what will be the value of base-X (in decimal) _____?

Q.21 Which of the following is not true?
(a) The r 's complement of a positive number N in base r is $(r^n - N)$.
(b) The $(r - 1)$'s complement of a positive number N in base r is $(r^n - N - 1)$.
(c) The $(r - 1)$'s complement of a positive number N having n digits and m digits in integer and fraction respectively in base r is $(r^n - r^m - N)$.
(d) The $(r - 1)$'s complement of a positive number N having n digit and m digits in integer and fraction part respectively in base r is $(r^n - r^m - N)$.

Q.22 The representation of the value of 20 bit signed integer in 2's complement form is $P = (A72E5)_{16}$. Which of the following represents $16 \times P$ in 1's complement representation?
(a) $(72E4F)_{16}$ (b) $(72E50)_{16}$
(c) $(72E4E)_{16}$ (d) None of the above

Q.23 If the value of $X + Y = 1$, then the value of $X \oplus Y$ is equal to
(a) X (b) $\bar{X} + \bar{Y}$
(c) 1 (d) 0

Q.24 Consider the circuit shown below:



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Then, which of the following statement is true.

- (a) When $A = 1$ and $B = 1$, Output $X = 0$ and $Y = 1$
(b) When $A = 1$ and $B = 0$, Output $X = 0$ and $Y = 1$
(c) When $A = 0$ and $B = 1$, Output $X = 1$ and $Y = 1$
(d) When $A = 0$ and $B = 0$, Output $X = 1$ and $Y = 0$

Q.25 The minimum decimal equivalent of the number $(1AC)_x$ is equal to _____.

Q.26 Consider, the Karnaugh map given below, where x represents "dont care" in function $f(w, x, y, z)$

wx	00	01	11	10
yz				
00	1			
01	1	1	x	
11		x	1	1
10				1

Total number of essential prime implicants are _____.

Q.27 If a Boolean function is having cyclic prime implicants K-map, then the number of minimal forms for function is _____.

Q.28 Given the following Karnaugh map, which one of the following represents the minimal sum of products of the map?

wx	00	01	11	10
yz				
00	0	$\times$	0	$\times$
01	$\times$	1	$\times$	1
11	0	$\times$	1	0
10	0	1	$\times$	0

- (a) $xy + \bar{y}z$ (b) $w\bar{x}\bar{y} + xy + xz$
(c) $\bar{w}x + \bar{y}z + xy$ (d) $xz + y$

Q.29 A logic circuit implements the Boolean function $F = \bar{X}Y + X\bar{Y}\bar{Z}$. It is found that the input combination $X = Y = 1$ can never occur. Taking this into account, a simplified expression for F is given by

- (a) $\bar{X} + \bar{Y}\bar{Z}$ (b) $X + Z$
(c) $X - Z$ (d) $Y + X\bar{Z}$

Q.30 Consider the Karnaugh map given below, where X represents "don't care" and blank represents 0.

		cd			
		00	01	11	10
ab	00		x	x	
	01	1			x
	11	x			1
	10		x	x	

Assume for all input (a, b, c, d) the respective complements $(\bar{a}, \bar{b}, \bar{c}, \bar{d})$ are also available. The above logic is implemented using 2-input NOR gates only. The minimum number of gates required is _____.

Q.31 Binary data is being represented in size of byte and in 2's complement form. The number of 1's present in representation of $(-59)_{10}$ is

- (a) 3 (b) 5
(c) 2 (d) 4

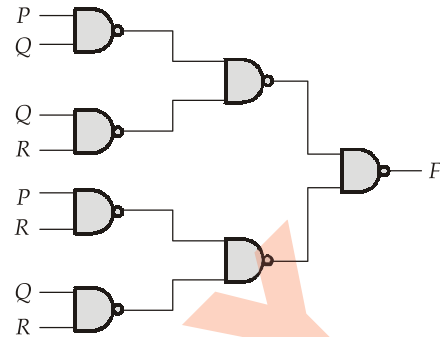
Q.32 The addition of the two numbers $(7FF)_{16} + (743)_8$ will be

- (a) $(8F2)_{16}$ (b) $(9E2)_{16}$
(c) $(F42)_{16}$ (d) $(EF2)_{16}$

Q.33 Which of the following expression is equivalent to $(A \oplus B) \oplus C$?

- (a) $(A + B + C)(\bar{A} + \bar{B} + \bar{C})$
(b) $(A + B + C)(\bar{A} + \bar{B} + C)$
(c) $ABC + \bar{A}(B \oplus C) + \bar{B}(A \oplus C)$
(d) None of these

Q.34 What is the Boolean expression for the output f of the combinational logic circuit of NAND gates given below:



- (a) $\bar{Q} + \bar{R}$ (b) $\bar{Q} \cdot \bar{R}$
(c) $\bar{P} + \bar{R}$ (d) $\bar{P} \cdot \bar{Q} \cdot \bar{R}$

Q.35 The minterm expansion of

$$f(P, Q, R) = PQ + Q\bar{R} + P\bar{R} \text{ is}$$

- (a) $m_2 + m_4 + m_6 + m_7$
(b) $m_0 + m_1 + m_3 + m_5$
(c) $m_0 + m_1 + m_6 + m_1$
(d) $m_2 + m_3 + m_4 + m_5$

Q.36 Let $A = 11111010$ and $B = 00001010$ be two 8 bit 2's complement numbers. Their product in 2's complement is

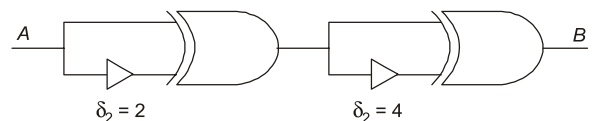
- (a) 11000100 (b) 10011100
(c) 10100101 (d) 11010101

Q.37 The number $(110011)_2$ is numerically equal to

- (a) $(51)_9$ (b) $(63)_8$
(c) $(33)_{16}$ (d) $(303)_4$ [MSQ]

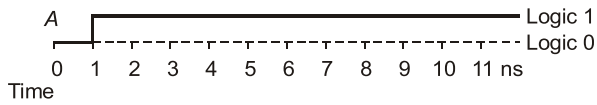
LEVEL 2 Questions

Q.38 Consider the following circuit composed of XOR gates and non-inverting buffers.



The non-inverting buffers have delays $\delta_1 = 2$ ns and $\delta_2 = 4$ ns as shown in the figure. Both XOR gates and all wires have zero delay. Assume that all gate inputs, outputs and wires are stable at logic level 0 at time 0. If the following waveform is applied at input A, how many

transition (s) (change of logic levels) occur (s) at B during the interval from 0 to 10 ns?



- (a) 1 (b) 2
(c) 3 (d) 4

Q.39 Consider a Boolean function $f(w, x, y, z)$. Suppose that exactly one of its inputs is allowed to change at a time. If the function happens to be true for two input vectors $i_1 = \langle w_1, x_1, y_1, z_1 \rangle$ and $i_2 = \langle w_2, x_2, y_2, z_2 \rangle$, we would like the function to remain true as the input changes from i_1 to i_2 (i_1 and i_2 differ in exactly one bit position), without becoming false momentarily. Let $f(w, x, y, z) = \Sigma(5, 7, 11, 12, 13, 15)$. Which of the following cube covers of f will ensure that the required property is satisfied?

- (a) $\bar{w}xz, wx\bar{y}, x\bar{y}z, xyz, wyz$
(b) $wxy, \bar{w}xz, wyz$
(c) $wx\bar{y}z, xz, w\bar{x}yz$
(d) $wyz, wxz, \bar{w}xz, x\bar{y}z, xyz$

Q.40 Consider numbers represented in 4-bit gray code. Let $h_3h_2h_1h_0$ be the gray code representation of a number n and let $g_3g_2g_1g_0$ be the gray code of $(n+1)$ (modulo 16) value of the number. Which one of the following functions is correct?

- (a) $g_0(h_3h_2h_1h_0) = \Sigma(1, 2, 3, 6, 10, 13, 14, 15)$
(b) $g_1(h_3h_2h_1h_0) = \Sigma(4, 9, 10, 11, 12, 13, 14, 15)$
(c) $g_2(h_3h_2h_1h_0) = \Sigma(2, 4, 5, 6, 7, 12, 13, 15)$
(d) $g_3(h_3h_2h_1h_0) = \Sigma(0, 1, 6, 7, 10, 11, 12, 13)$

Q.41 Consider the operations

$$f(X, Y, Z) = X'YZ + XY' + Y'Z'$$

$$g(X, Y, Z) = X'YZ + X'YZ' + XY$$

Which one of the following is correct?

- (a) Both $\{f\}$ and $\{g\}$ are functionally complete
(b) Only $\{f\}$ is functionally complete
(c) Only $\{g\}$ is functionally complete
(d) Neither $\{f\}$ nor $\{g\}$ is functionally complete

Q.42 Which one of the following statement is true about the cyclic prime implicant K-map function?

- (a) The function is having two minimal forms, with one common prime implicant.
(b) The function is having two minimal forms, with two common prime implicants.
(c) The function is having two minimal forms, with no common prime implicants.
(d) The function is having two minimal forms and one essential prime implicant.

Q.43 A lawn-sprinkling system is controlled automatically by certain combination of following variables.

Season ($S = 1$, if summer ; 0, otherwise)

Moisture content of the soil ($M = 1$, if high ; 0, if low)

Outside temperature ($T = 1$, if high ; 0, if low)

Outside humidity ($H = 1$, if high ; 0, if low)

The sprinkler is turned 'ON' under any of the following conditions.

1. The moisture content is low in winter.
2. The temperature is high and moisture content is low in summer.
3. The temperature and humidity is high in summer.
4. The temperature and moisture content is low in summer.
5. The temperature is high and the humidity is low.

Then the minimum number of two input NAND gates required to implement this function is _____.

Q.44 For the map shown below, the minimized logical expression in sum of product (SOP) form is

CD \ AB	00	01	10	11
00	1		1	1
01				
10		1	1	
11	1	1	1	1

- (a) $\bar{B}\bar{D} + AD + AC$
 (b) $AB + AD + \bar{B}\bar{D}$
 (c) $AB + A\bar{C}D + AC\bar{D} + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{D}$
 (d) $\bar{A}\bar{B}\bar{C} + AC + \bar{A}\bar{B}\bar{D}$

Q.45 Consider the following Boolean function:

$$Y = (B + C)(\bar{A} + \bar{D})$$

The minimum number of 2 input NAND gates required to implement the above function is

- (a) 3 (b) 4
 (c) 5 (d) 6

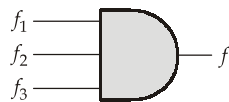
Q.46 The minimum number of 2-input NOR gates required to implement the Boolean function

$$F = \bar{X}Z + \bar{X}W + \bar{Y}Z + \bar{Y}W \text{ is } \underline{\hspace{2cm}}.$$

Q.47 Consider the logical functions given below:

$$f_1(A, B, C) = \Sigma(2, 3, 4)$$

$$f_2(A, B, C) = \pi(0, 1, 3, 6, 7)$$



If f is logic zero, then maximum number of possible minterms in function f_3 are _____.

Q.48 The minimal function that can detect a 'divisible' by 3' 8421 BCD code digit (representation is $D_8 D_4 D_2 D_1$) is given by

- (a) $D_8 D_1 + D_4 D_2 + D_8 D_2 D_1$
 (b) $D_4 D_1 + D_4 D_2 + D_8 \bar{D}_4 \bar{D}_2 D_1$
 (c) $D_8 D_1 + D_4 D_2 \bar{D}_1 + \bar{D}_4 D_2 D_1 + \bar{D}_8 \bar{D}_4 \bar{D}_2 \bar{D}_1$
 (d) $D_4 D_2 \bar{D}_1 + \bar{D}_4 D_2 D_1 + D_8 D_4 D_2 D_1$

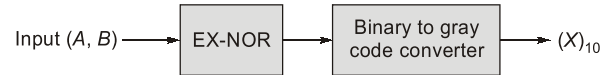
Q.49 The maximum number of Boolean expressions that can be formed for the function $f(x, y, z)$ satisfying the relation $f(\bar{x}, y, \bar{z}) = f(x, y, z)$ is _____.

Q.50 Simplify the function $F = \Sigma m(1, 2, 3, 4, 5, 8, 9, 10)$. The number of literals in the minimal product of sum form are _____.

Q.51 The number of minterms after minimizing the following Boolean expression is _____.

$$[D' + AB' + A'C + AC'D + A'C'D]'$$

Q.52 Consider the input $A = 10110100$ and $B = 01110000$ is fed as input as shown in the below diagram:



The value of X is _____.

Q.53 Which of the following is true?

- (a) Number of Prime Implicants (PI) is same as the number of terms obtained in minimized function of K-map.
 (b) Minterms of K-map is always the subset of Prime Implicants (PI).
 (c) Both are true
 (d) None of the above

Q.54 Given the function $F = X + Y'Z$, when F is a function of 3 Boolean variables X, Y and Z . Consider the following statements:

$$S_1 : F = \Sigma(0, 1, 2, 3, 4)$$

$$S_2 : F = \Sigma(1, 4, 5, 6, 7)$$

$$S_3 : F = \Pi(0, 2, 3)$$

$$S_4 : F = \Pi(5, 6, 7)$$

Which of the following is true?

- (a) S_1 - false, S_2 - true, S_3 - false, S_4 - true
 (b) S_1 - true, S_2 - false, S_3 - true, S_4 - false
 (c) S_1 - true, S_2 - false, S_3 - false, S_4 - true
 (d) S_1 - false, S_2 - true, S_3 - true, S_4 - false

Q.55 Let $A = 11111010$ and $B = 00001111$ be two 8-bit 2's complement numbers. Their product in 2's complements is

- (a) 01011010 (b) 10100110
 (c) 10010010 (d) 11010101

Q.56 If $Y = ABC + \bar{B}C + \bar{A}C + \bar{A}\bar{C}$ then dual of Y are

- (a) $(A + B + C) \cdot [\bar{B}\bar{C} + A] \cdot (A\bar{C})$
 (b) $(A + B + C) \cdot [(B + C) \cdot (C)] \cdot (\bar{A} + \bar{C})$
 (c) $(A + B + C) \cdot (\bar{B}\bar{C} + A\bar{C}) \cdot (\bar{A} + \bar{C})$
 (d) $(ABC) \cdot (\bar{B}\bar{C} + \bar{A}C) \cdot (\bar{A} + \bar{C})$

Q.57 How many minimum number of decimal digits is required to represent 19 bit of binary data. The number of decimal digit will be _____.

Q.58 Let A, B and C are Boolean variables, then which of the following is true?

- (a) $(\overline{AB} + \overline{C})(A + C) = (A + \overline{B})C$
- (b) $(A + \overline{B} + \overline{C})(A + \overline{BC}) = \overline{A}(B + \overline{C})$
- (c) $(A + \overline{AB}\overline{C})B + \overline{AB} = BA + \overline{AB}$
- (d) All of the above

Q.59 Let $X = X_2X_1X_0$ and $Y = Y_1Y_0$ be unsigned positive 3-digit and 2-digit numbers respectively. The output function 'f' = 1 only when $X > Y$ otherwise '0'. Then the value of output f is equal to

- (a) $(X_2 + Y_1 + Y_0)(X_2 + \overline{X}_1 + \overline{Y}_0)(X_2 + X_1 + \overline{Y}_1)(X_2 + X_1 + X_0)(X_2 + X_0 + \overline{Y}_1)$
- (b) $(X_2 + \overline{Y}_1 + \overline{Y}_0)(X_2 + X_1 + \overline{Y}_0)(X_2 + \overline{X}_1 + Y_1)(X_2 + X_1 + X_0)(X_2 + X_0 + \overline{Y}_1)$
- (c) $(X_2 + \overline{Y}_1 + \overline{Y}_0)(X_2 + X_1 + \overline{Y}_0)(X_2 + X_1 + \overline{Y}_1)(X_2 + X_1 + X_0)(X_2 + X_0 + \overline{Y}_1)$
- (d) $(X_2 + \overline{Y}_1 + \overline{Y}_0)(X_2 + \overline{X}_1 + \overline{Y}_0)(X_2 + X_1 + \overline{Y}_1)(X_2 + \overline{X}_1 + \overline{X}_0)(X_2 + X_0 + \overline{Y}_1)$

Q.60 Consider the function $F = A(\overline{A} + B)(\overline{A} + B + \overline{C})$, where F is a function in three Boolean variables A, B and C and $\overline{A}, \overline{C}$ are complement of variable A and C. Consider the following statements:

- S_1 : $F = \Sigma(6, 7)$
- S_2 : $F = \Sigma(0, 1, 2, 3, 4, 5)$
- S_3 : $F = \Pi(0, 1, 2, 3, 4, 5)$
- S_4 : $F = \Pi(6, 7)$

Which of the following is true?

- (a) $S_1 = \text{true}, S_2 = \text{false}, S_3 = \text{false}, S_4 = \text{true}$
- (b) $S_1 = \text{false}, S_2 = \text{true}, S_3 = \text{false}, S_4 = \text{true}$
- (c) $S_1 = \text{true}, S_2 = \text{false}, S_3 = \text{true}, S_4 = \text{false}$
- (d) $S_1 = \text{false}, S_2 = \text{true}, S_3 = \text{true}, S_4 = \text{false}$

Q.61 Consider the following arithmetic equation:

$$\frac{302}{20} = 12.1$$

The minimum possible non-zero base for the given system is _____.

Q.62 The number of minterms after minimizing the following Boolean expression is _____.

$$[D' + AB' + A'C + AC'D + A'C'D]'$$

Q.63 The simultaneous equations on the Boolean variables x, y, z and w

$$y + z + w = 1$$

$$yz = 0$$

$$yw + x = 1$$

$$yz + \overline{w} \cdot \overline{x} = 0$$

have the following solution for x, y, z and w, respectively

- (a) 1000
- (b) 1011
- (c) 0111
- (d) 0001

Q.64 Let $\oplus$ and $\odot$ denote the exclusive OR and exclusive NOR operations, respectively which one of the following is NOT correct?

- (a) $\overline{P} \oplus \overline{Q} = P \oplus Q$
- (b) $(P \oplus \overline{P}) \oplus \overline{Q} = (P \odot \overline{P}) \odot \overline{Q}$
- (c) $\overline{P} \oplus Q = \overline{P} \odot \overline{Q}$
- (d) $(P \oplus \overline{P}) \oplus (Q \oplus \overline{Q}) = (P \odot \overline{P}) \odot (Q \odot \overline{Q})$

Q.65 Consider the minterm list form of a Boolean function F given below:

$$F(P, Q, R, S) = \Sigma m(0, 2, 5, 7, 9, 13) + d(3, 8, 10, 12)$$

Here m denotes a minterm and d denotes a don't care term. The number of essential prime implicants of the function F is _____.

Q.66 Number of pulses needed to change the contents of a 8-bit up counter from 10101100 to 00100111 (right most bit is the LSB) are _____

Q.67 If $(7.FD6)_{16} = (7.7726)_x$ and $(7864)_{10} = (1EB8)_y$, Then the values of x and y are respectively

- (a) 4 and 8
- (b) 10 and 16
- (c) 8 and 16
- (d) 8 and 8

Q.68 The minimized expression for the Boolean function $f(A, B, C, D) = \Sigma m(0, 2, 3, 4, 6, 8, 10, 11)$

- (a) $\overline{A}\overline{D} + \overline{B}D + \overline{B}C$
- (b) $\overline{A}\overline{D} + B\overline{D} + B\overline{C}$
- (c) $\overline{A}\overline{D} + \overline{B}\overline{D} + BC$
- (d) $\overline{A}\overline{D} + \overline{B}\overline{D} + \overline{B}C$

Q.69 Consider the addition of numbers with different bases,

$$(X)_7 + (Y)_8 + (W)_{10} + (Z) = (K)_9$$

If $X = 36, Y = 67, W = 98$ and $K = 241$ then Z is equal to

- (a) $(19)_{10}$
- (b) $(34)_5$
- (c) $(13)_{16}$
- (d) $(34)_{10}$

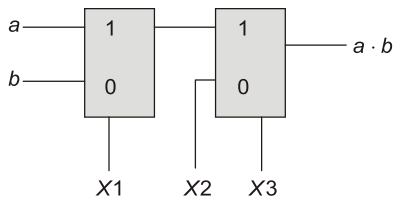
[MSQ]



Combinational Circuits

LEVEL 1 Questions

Q.1 The following circuit implements a two-input AND gate using two 2-1 multiplexers.



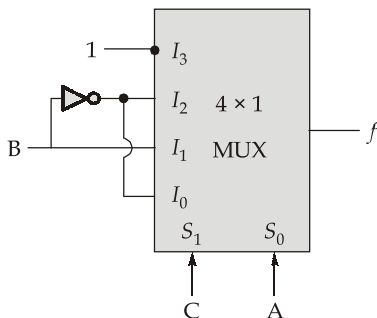
What are the values of $X1$, $X2$, $X3$?

- (a) $X1 = b$, $X2 = 0$, $X3 = a$
- (b) $X1 = b$, $X2 = 1$, $X3 = b$
- (c) $X1 = a$, $X2 = b$, $X3 = 1$
- (d) $X1 = a$, $X2 = 0$, $X3 = b$

Q.2 When two 8-bit numbers $A_7 \dots A_0$ and $B_7 \dots B_0$ in 2's complement representation (with A_0 and B_0 as the least significant bits) are added using a **ripple-carry adder**, the sum bits obtained are $S_7 \dots S_0$ and the carry bits are $C_7 \dots C_0$. An overflow is said to have occurred if

- (a) the carry bit C_7 is 1
- (b) all the carry bits ($C_7, \dots, C_0$) are 1
- (c) $(A_7 \cdot B_7 \cdot \bar{S}_7 + \bar{A}_7 \cdot \bar{B}_7 \cdot S_7)$ is 1
- (d) $(A_0 \cdot B_0 \cdot \bar{S}_0 + \bar{A}_0 \cdot \bar{B}_0 \cdot S_0)$ is 1

Q.3 Find the function $f(A, B, C)$ realized by the given 4×1 MUX.

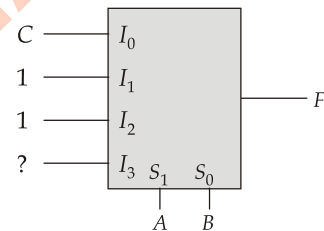


- (a) $\Sigma m(0, 1, 2, 4)$
- (b) $\Sigma m(0, 1, 5, 6, 7)$
- (c) $\Sigma m(0, 1, 2, 4, 5)$
- (d) $\Sigma m(2, 4, 5, 6, 7)$

Q.4 Which of the following is not correct about the multiplexer circuit?

- (a) A MUX circuit can be used as a universal logic gate.
- (b) A MUX has 2^n input lines and 1 output line.
- (c) A 16×1 MUX can be designed using 2×1 MUX.
- (d) None of the above

Q.5 For the 4×1 MUX shown below:



If the output $F = A + B + C$, then I_3 will be

- (a) A
- (b) 0
- (c) C
- (d) All of the above

Q.6 Consider the below digital circuit. The output f of both the circuit is same.

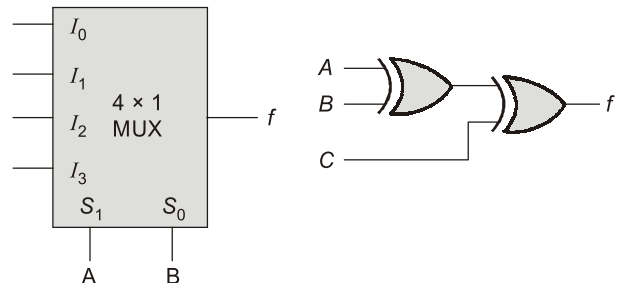


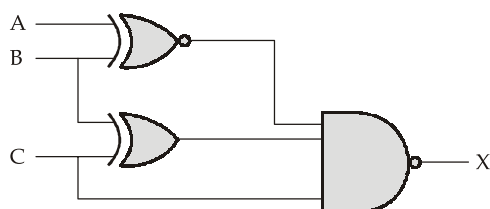
Figure (a)

Figure (b)

What are the correct combinations of inputs of multiplexer using the codes given below?

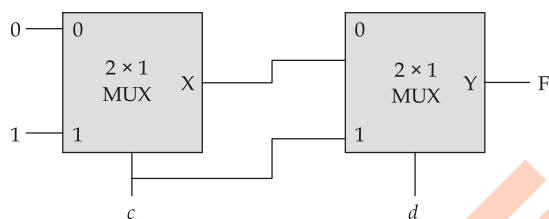
	I_0	I_1	I_2	I_3
(a)	C	$\bar{C}$	$\bar{C}$	C
(b)	0	1	C	$\bar{C}$
(c)	1	$\bar{C}$	$\bar{C}$	C
(d)	C	0	1	C

Q.7 Find the correct input condition (ABC) which produces $X = 0$ in the logic circuit shown below:



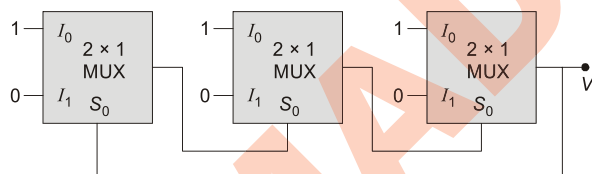
- (a) 111 (b) 101
(c) 100 (d) 001

Q.8 Consider the below digital circuit.



- (a) $c \oplus d$ (b) $c \odot d$
(c) cd (d) c

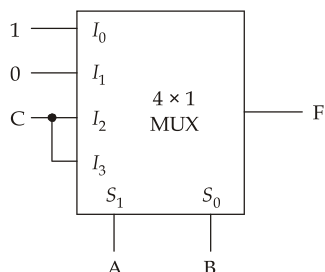
Q.9 Consider the below circuit:



The propagation delay of each multiplexer is 50 ns. The frequency of the output signal V is

- (a) 4 MHz (b) 3.33 MHz
(c) 5 MHz (d) 5.33 MHz

Q.10 The number of minterms of the following function F is _____.



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Q.11 Consider a 4-bit ripple carry adder for computing sum of 2 different numbers X and Y. These numbers are represented in 2's complement form. If the decimal value of X is +7 the minimal decimal value of Y that leads to longest latency for the sum to stabilize is _____.

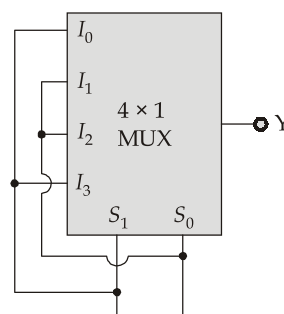
Q.12 If Booth's Algorithm for multiplication is used then which of the following represents multiplier -29 in recorded form?

- (a) $00 - 10 + 100 - 1$ (b) $00 - 100 - 10 + 1$
(c) $00 - 100 + 10 - 1$ (d) $00 - 100 + 10 + 1$

Q.13 The minimum number of 2×1 MUX required to implement a half-subtractor circuit when only basic inputs 0, 1, A and B are available is

- (a) 3 (b) 4
(c) 5 (d) 6

Q.14 A gate have two inputs (A, B) and one output (Y) is implemented using a 4×1 MUX as shown in the figure below:

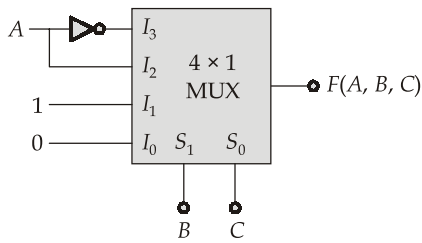


If the function 'Y' = B then the select line will be:

- (a) $S_0 = B, S_1 = A$ (b) $S_0 = A, S_1 = \bar{B}$
(c) $S_0 = \bar{A}, S_1 = B$ (d) $S_0 = \bar{A}, S_1 = \bar{B}$

Q.15 The total number of NAND gates required to implement a 4×1 multiplexer is (assuming a NAND gates of any number of inputs are available) _____.

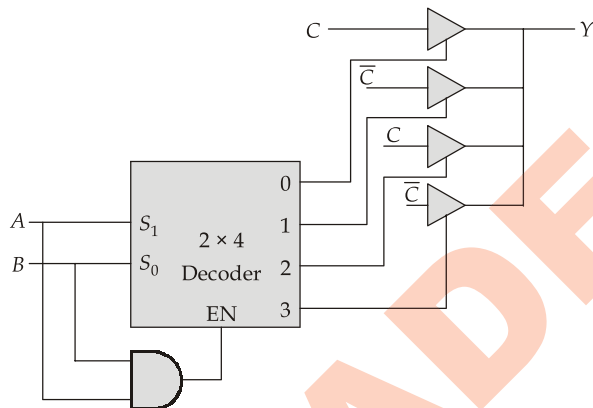
Q.16 A 4×1 MUX is used to implement a 3-input Boolean function as shown in figure. The Boolean function $F(A, B, C)$ implemented is



- (a) $F(A, B, C) = \sum m(3, 5, 6)$
 (b) $F(A, B, C) = \sum m(1, 5, 6)$
 (c) $F(A, B, C) = \sum m(2, 4, 5, 6)$
 (d) $F(A, B, C) = \sum m(1, 3, 5, 6)$

Q.17 Number of 3 to 8 line decoders with an enable input are needed to construct a 9 to 512 line decoder without using any other logic gates. Then the value of P is _____.

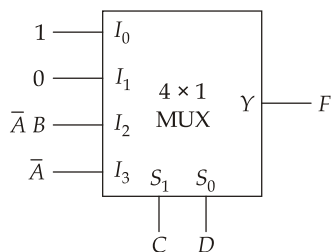
Q.18 Consider the logic diagram consisting of tristate buffers shown below:



Then the output (Y) of the logic diagram is equal to

- (a) C (b) $\bar{C}$
 (c) ABC (d) $AB\bar{C}$

Q.19 Consider a 4×1 MUX circuit as shown below:

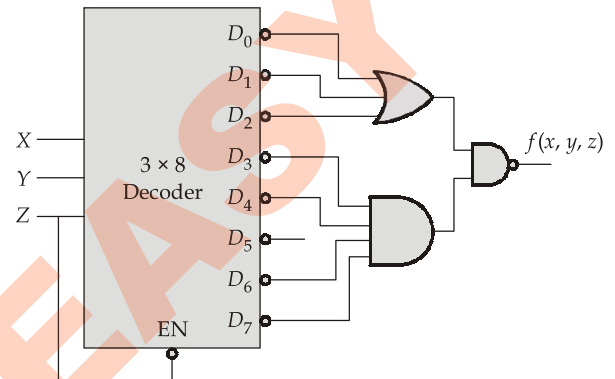


Then the output, ' F ' is equal to

- (a) $\bar{A}CD + \bar{A}BC + \bar{A}\bar{C}\bar{D} + A\bar{C}\bar{D}$
 (b) $\bar{A}CD + \bar{A}BC + \bar{C}\bar{D}$
 (c) $\bar{A}CD + \bar{A}BC + A\bar{C}\bar{D} + A\bar{C}\bar{D}$
 (d) $\bar{A}\bar{C}\bar{D} + \bar{A}\bar{B}C + A\bar{C}\bar{D} + ACD$

[MSQ]

Q.20 The Boolean expression $f(x, y, z)$ in its canonical form for the decoder circuit shown below:

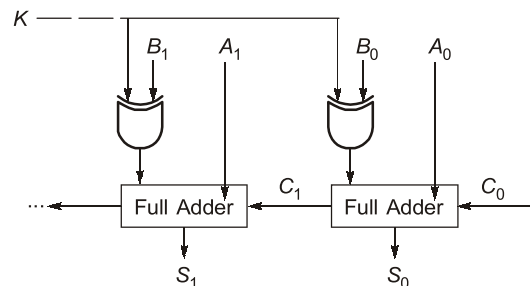


- (a) $\pi M(4, 6)$
 (b) $\pi M(0, 1, 2, 3, 5, 7)$
 (c) $\sum m(0, 1, 2, 3, 5, 7)$
 (d) $\sum m(4, 6)$

[MSQ]

LEVEL 2 Questions

Q.21 Consider the ALU shown below:



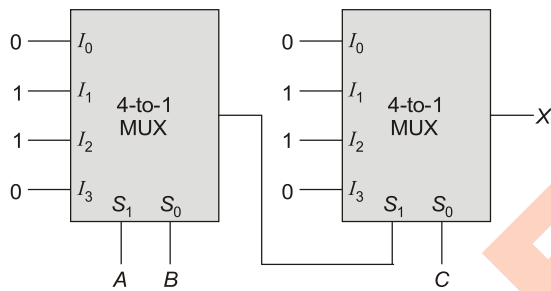
If the operands are in 2's complement representation, which of the following operations can be performed by suitably setting the control lines K and C_0 only (+ and - denote addition and subtraction respectively)?

- (a) $A + B$, and $A - B$, but not $A + 1$
- (b) $A + B$, and $A + 1$, but not $A - B$
- (c) $A + B$, but not $A - B$, or $A + 1$
- (d) $A + B$, and $A - B$, and $A + 1$

Q.22 A half adder is implemented with XOR and AND gates. A full adder is implemented with two half adders and one OR gate. The propagation delay of an XOR gate is twice that of an AND/OR gate. The propagation delay of an AND/OR gate is 1.2 microseconds.

A 4-bit ripple-carry binary adder is implemented by using four full adders. The total propagation time of this 4-bit binary adder in microseconds is _____.

Q.23 In the following circuit, X is given by



- (a) $X = A\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}C + ABC$
- (b) $X = \bar{A}BC + A\bar{B}C + ABC + \bar{A}\bar{B}\bar{C}$
- (c) $X = AB + BC + AC$
- (d) $X = \bar{A}\bar{B} + \bar{B}\bar{C} + \bar{A}\bar{C}$

Q.24 Figure I shows a 4-bit ripple carry adder realized using full adders and Figure II shows the circuit of a full-adder (FA). The propagation delay of the XOR, AND and OR gates in Figure II are 20 ns, 15 ns and 10 ns, respectively. Assume all the inputs to the 4-bit adder are initially reset to 0.

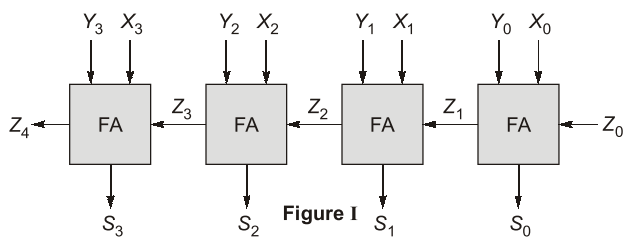


Figure I

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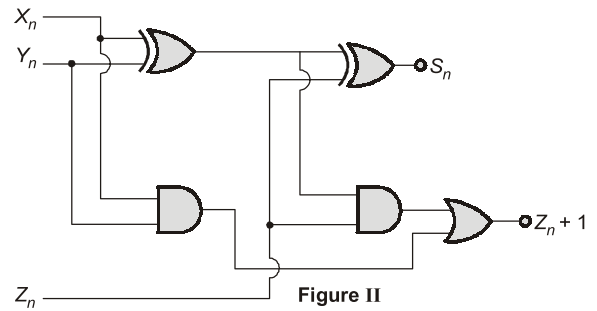
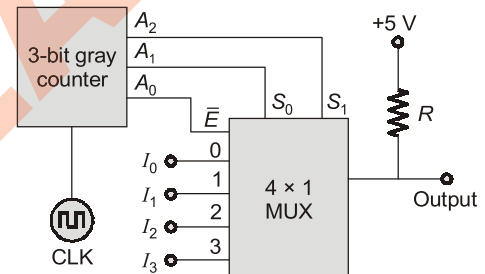


Figure II

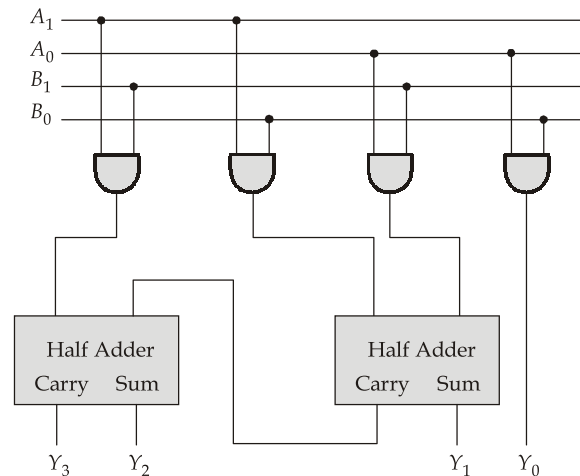
At $t = 0$, the inputs to the 4-bit adder are changed to $X_3X_2X_1X_0 = 1100$, $Y_3Y_2Y_1Y_0 = 0100$ and $Z_0 = 1$. The output of the ripple carry adder will be stable at t (in ns) = _____.

Q.25 A 3-bit gray counter is used to control the output of the multiplexer as shown in the figure. The initial state of the counter is 000_2 . The output is pulled high. The output of the circuit follows the sequence



- (a) $I_0, 1, 1, I_1, I_3, 1, 1, I_2$
- (b) $I_0, 1, I_1, 1, I_2, 1, I_2, 1$
- (c) $1, I_0, 1, I_1, I_2, 1, I_3, 1$
- (d) $I_0, I_1, I_2, I_3, I_0, I_1, I_2, I_3$

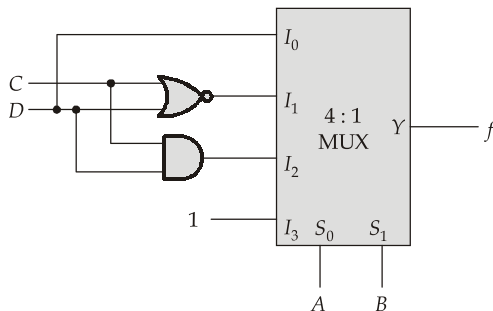
Q.26 Consider the circuit shown in the figure below:



The input to the circuit is two, 2-bit numbers. The numbers are represented as (A_1A_0) and (B_1B_0) . The function of the circuit is

- (a) A look ahead carry adder circuit
- (b) An array divider circuit
- (c) A parity check circuit
- (d) An array multiplier circuit

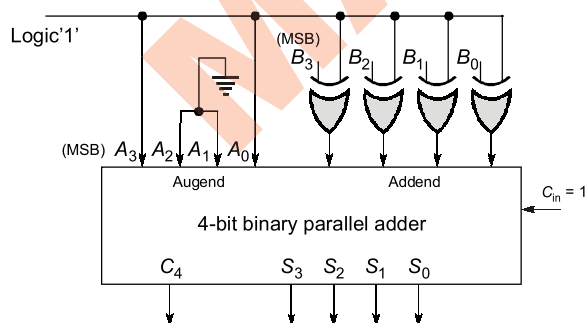
Q.27 Consider the circuit shown in the figure below:



If A and B are connected to the select lines of the MUX circuit, then the Boolean function realized by the circuit is

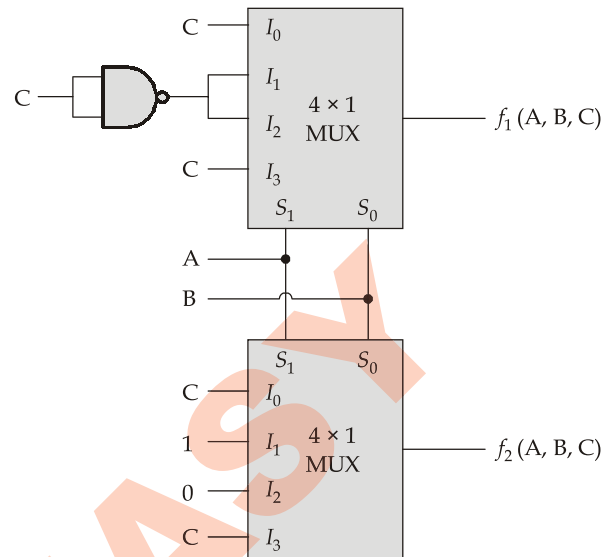
- (a) $f(A, B, C, D) = \Pi M(1, 3, 7, 8, 12, 13, 14, 15)$
- (b) $f(A, B, C, D) = \Sigma m(1, 3, 7, 9, 12, 13, 14, 15)$
- (c) $f(A, B, C, D) = \Pi M(0, 2, 4, 5, 6, 9, 10, 11)$
- (d) $f(A, B, C, D) = \Sigma m(0, 1, 4, 5, 6, 9, 10, 11)$

Q.28 Consider the digital circuit shown below, single digit number (B) is converted into its 4 bit binary equivalent ($B_3B_2B_1B_0$) and then applied to the addend bits of the adder as shown below:



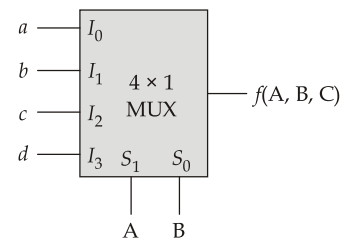
If number B is 15 then the output $S_3 S_2 S_1 S_0$ is _____.

Q.29 The multiplexer circuit shown below will function as a

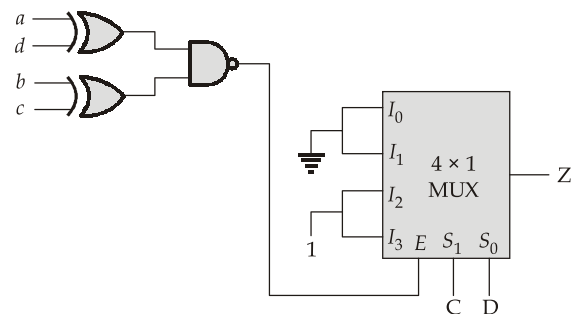


- (a) Full adder
- (b) Full subtractor
- (c) Half adder
- (d) Half subtractor

Q.30 Given $f(A, B, C) = \Sigma m(0, 1, 4, 5)$ for the given 4×1 MUX

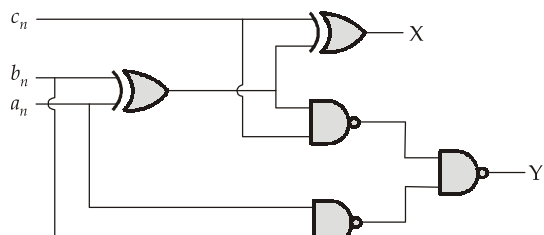


Then the output Z of 4×1 MUX will be



- (a) 0
- (b) $C\bar{D}$
- (c) CD
- (d) 1

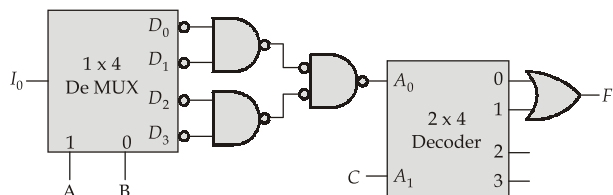
Q.31 A circuit is shown below:



The propagation time for each NAND gate is 10 nsec. What is the time required to get the output Y (in nsec). (Assume that all other gates are implemented using minimum number of NAND gates)

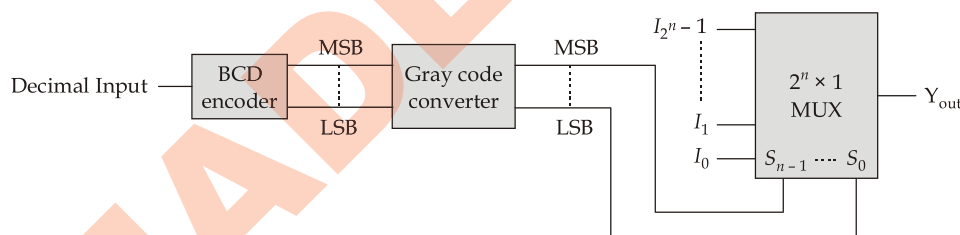
- (a) 20 (b) 40
(c) 50 (d) 60

Q.32 Consider the logic circuit given below:



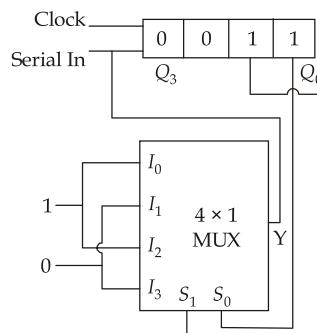
The minimized expression for F is

Q.35 Consider the circuit given below:



If the decimal input is 89 then Y_{out} corresponds to I_m , then the value of m is _____.

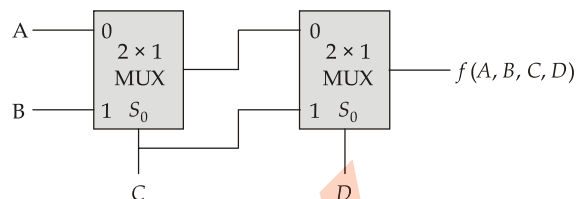
Q.36 The initial content of serial IN parallel OUT, right shift, shift register shown below is 0011. After how many clock pulses, the content of register will return to its initial value.



- (a) 7 (b) 8 (c) 11 (d) 13

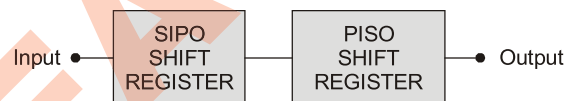
- (a) $\bar{C}$ (b) I_0
(c) C (d) $\bar{I}_0$

Q.33 Consider the logic circuit given below. The min terms in $f(A, B, C, D)$ are



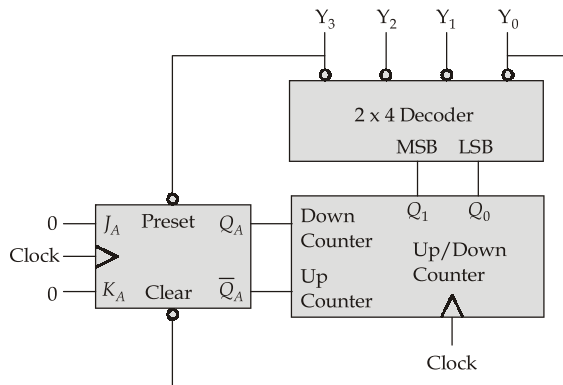
- (a) $\Sigma m(1, 3, 5, 6, 7, 11, 14)$
 (b) $\Sigma m(3, 6, 7, 8, 10, 12, 14, 15)$
 (c) $\Sigma m(3, 6, 7, 8, 11, 12, 14, 15)$
 (d) $\Sigma m(3, 6, 7, 9, 11, 12, 14, 15)$

Q.34 Two 4-bit shift register are cascaded as shown in below figure.



A 4-bit input 1001 is applied, then after how many clock pulses the same 4-bit data will appear at the output _____.

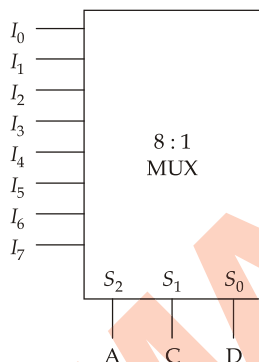
Q.37 Consider the circuit given below:



Assuming the initial value of counter output (Q_1, Q_0) as zero, the counter output for 8 clock pulses in decimal form is

- (a) 0, 3, 2, 1, 0, 1, 2, 3
- (b) 0, 1, 2, 3, 2, 1, 0, 1
- (c) 0, 1, 2, 3, 3, 2, 1, 0
- (d) 0, 3, 2, 1, 0, 0, 1, 2

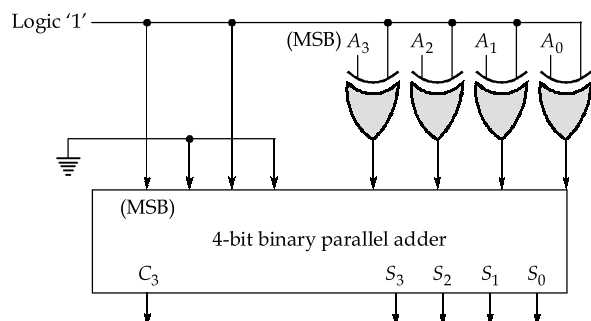
Q.38 The logic function $F(A, B, C, D) = AD + ABD + ABC + ACD$ is to be realized using an 8 to 1 multiplexer shown in the figure.



The input I_6 to the multiplexer will be

- (a) 0
- (b) 1
- (c) $\bar{B}$
- (d) B

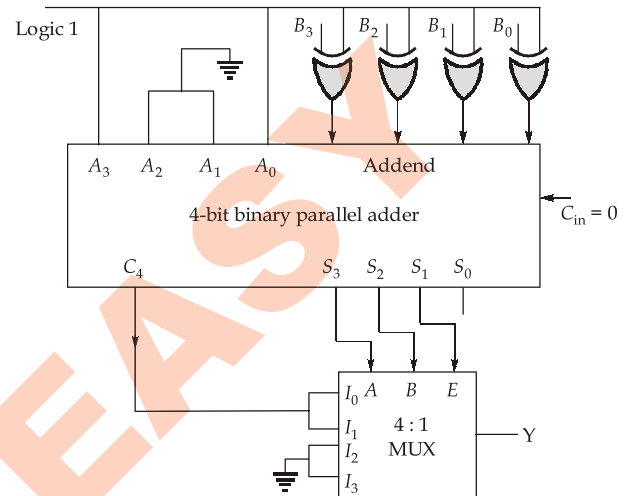
Q.39 Consider the digital circuit shown below:



The digital circuit acts as

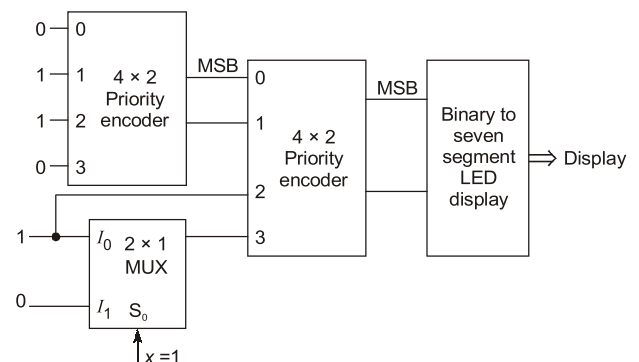
- (a) 9's complement circuit
- (b) 0's complement circuit
- (c) 5's complement circuit
- (d) 9's complement circuit if $C_3 = 0$ and 10's complement circuit if $C_3 = 1$.

Q.40 Consider the digital circuit shown below:



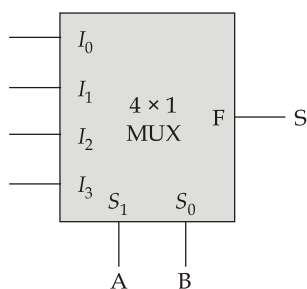
What will be the output Y, if the number $B_3 B_2 B_1 B_0 = 0101$ _____.

Q.41 Consider the combinational circuit below:



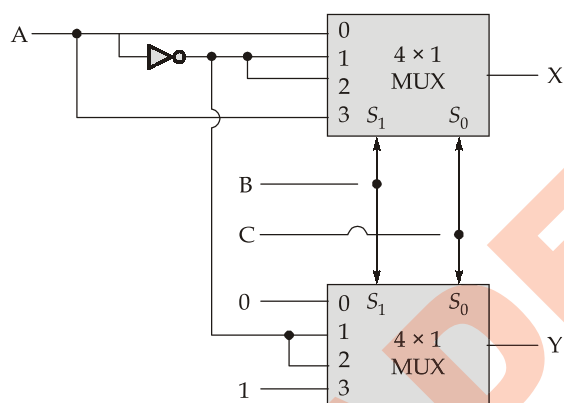
The output of the combinational circuit is _____.

Q.42 Consider a 4×1 MUX to be used to implement the sum of a 1-bit full adder with input bits A and B and the carry input C_{in} . Which of the following combination of inputs to I_0, I_1, I_2 and I_3 of the MUX will realize the sum S?



- (a) $I_0 = I_3 = C_{in}; I_1 = I_2 = \bar{C}_{in}$
 (b) $I_0 = I_3 = \bar{C}_{in}; I_1 = I_2 = C_{in}$
 (c) $I_0 = I_1 = C_{in}; I_2 = I_3 = \bar{C}_{in}$
 (d) $I_0 = I_1 = \bar{C}_{in}; I_2 = I_3 = C_{in}$

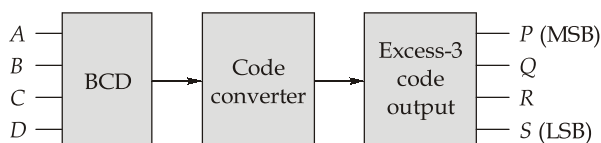
Q.43 Consider the circuit given below:



The circuit act as

- (a) Full adder
 (b) Full subtractor
 (c) Priority encoder
 (d) Half adder sum and half subtracter difference

Q.44 Consider a code converter as shown below that converts BCD to excess-3 code for the decimal digits.

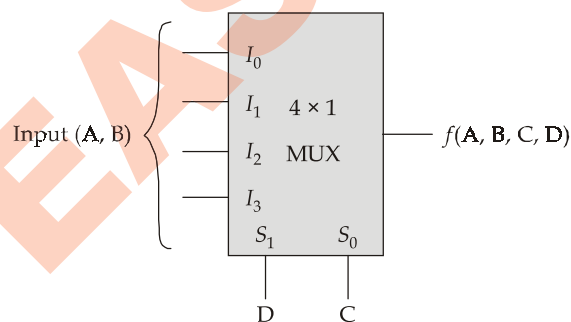


The simplified Boolean function output excess-3 code for R will be

- (a) $AB + (\bar{C} + \bar{D})$ (b) $C \oplus D$
 (c) $C \odot D$ (d) $AB + (\bar{C} \odot \bar{D})$

Q.45 A 16×1 MUX is implemented using 4×1 MUX. Assume at level 1 and level 2 of circuit X and Y number of MUX required respectively. Then value of $(X - Y)$ is _____.

Q.46 A 4×1 MUX is used to realize a 4-variable function, $f(A, B, C, D) = \sum m(0, 3, 4, 8, 9, 15)$. The inputs to the MUX are from variables A and B, then the input to the MUX from I_0 to I_3 respectively are



- (a) $\bar{A} + \bar{B}, 0, A\bar{B}, A \odot B$
 (b) $\bar{A}, 0, \bar{A}B, A \oplus B$
 (c) $\bar{A}\bar{B}, 0, A\bar{B}, A \odot B$
 (d) $\bar{A} + \bar{B}, 1, A\bar{B}, A \oplus B$

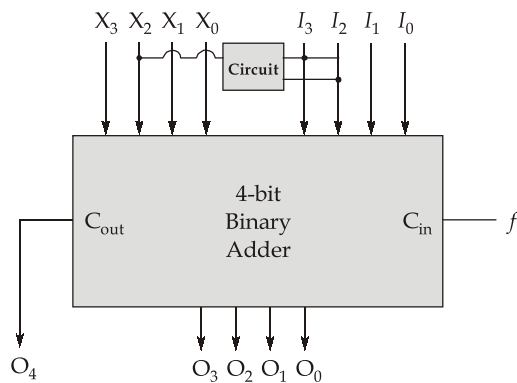
Q.47 A 1-bit binary comparator is designed. The only one output is active high for any given input to the comparator out of the 3 outputs. The variable X is used to count minimum number of minterms for equality checking, Y for greater than and Z for less than comparison then $X + Y + Z$ is

- (a) 3 (b) 4
 (c) 5 (d) 6

Q.48 Consider the below circuit which has four bit binary number as input and a five bit binary number as output:

Input: $I_3 I_2 I_1 I_0$
Output: $O_4 O_3 O_2 O_1 O_0$

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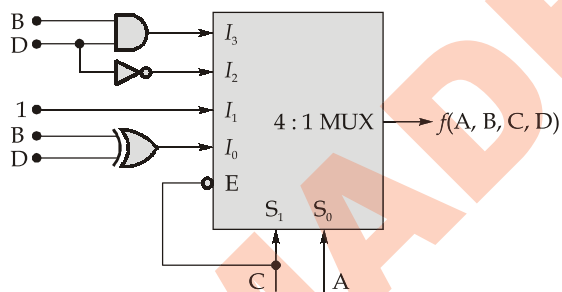


Which logic gate is required at circuit and what decimal value will be added at $(X_3X_2X_1X_0)$ to make this circuit binary to radix-12 conversion?

- (a) NAND, 2 (b) AND, 4
(c) NAND, 4 (d) AND, 12

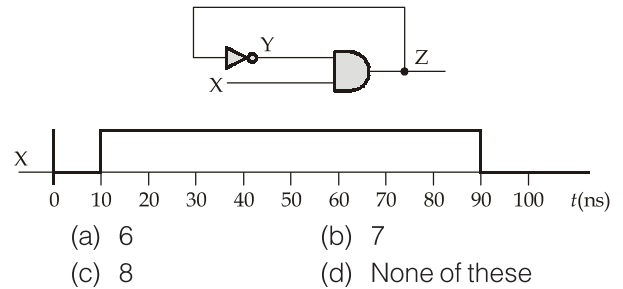
Q.49 A 3×8 decoder has used to implement a full adder and only 2-input OR-gate is available to get the sum and carry outputs. The number of OR-gates required is _____.

Q.50 Identify the function $f(A, B, C, D)$, which is realised by the given 4×1 multiplexer circuit.



- (a) $f = \sum m(0, 4, 8, 9, 12, 13)$
(b) $f = \sum m(1, 4, 8, 9, 12, 13)$
(c) $f = \sum m(1, 4, 8, 9, 12, 15)$
(d) None of the above

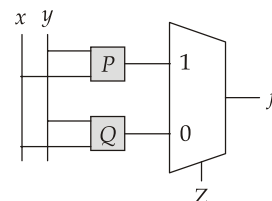
Q.51 Assume that the inverter in the network below has a propagation delay of 5 ns and the AND gate has a propagation delay of 10 ns. Draw a timing diagram for the network showing X, Y and Z. Assume Y is initially 1 and X waveform is given. Then the number of points where level changes is (excluding at $t = 0$ ns if any) at output Z till $t = 100$ ns is



Q.52 Consider a 3-bit number A and 2 bit number B are given to a multiplier. The output of multiplier is realized using AND gate and one bit full adders. If minimum number of AND gates required are X and one bit full adders required are Y, then $X + Y =$ _____.

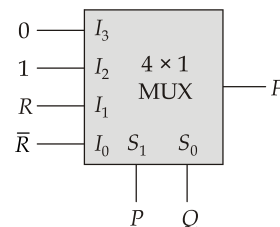


Q.53 The majority function is a Boolean function $f(x, y, z)$, that takes the value 1 whenever a majority of variables x, y, z are 1. In the circuit diagram for the majority function shown below. The logic gates for the boxes labeled P and Q are, respectively



- (a) AND, OR (b) OR, AND
(c) EXOR, OR (d) EXOR, AND

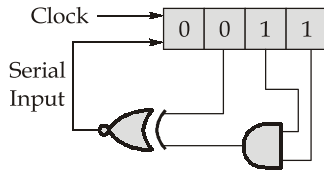
Q.54 Consider the 4 to 1 multiplexer with two select lines S_1 and S_0 gives below,



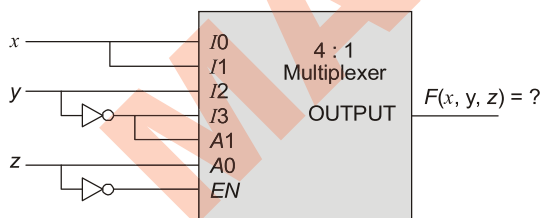
The minimal sum of products form of the Boolean expression for the output F of the multiplexer is

- (a) $\bar{P}QR + \bar{Q}\bar{R} + QP$ (b) $\bar{P}QR + Q\bar{P} + \bar{Q}\bar{R}$
(c) $P\bar{Q}R + \bar{Q}\bar{R} + \bar{Q}P$ (d) $\bar{Q}\bar{R} + \bar{Q}P + \bar{P}QR$

- Q.55** The shift register shown in figure is initially loaded with the bit pattern 0011. Subsequently the shift register is clocked and with each clock pulse the pattern gets shifted by one bit position to the right. With each shift, the bit at the serial input is pushed to the left most position (MSB). Clock pulses taken by the shift register to get the content 0011 again are _____



- Q.56** A half adder is implemented with XOR and AND gates. A full adder is implemented with two half adders and one OR gate. The propagation delay of an XOR gate is $1.5 \mu\text{s}$. The propagation delay of an AND/OR gate is $1 \mu\text{s}$. A 4-bit ripple carry binary adder is implemented by using 4 full adders. The total propagation time of this 4-bit binary adder (in μs) is _____. (Answer upto one decimal place)
- Q.57** Consider the following multiplexer where I_0, I_1, I_2, I_3 are four data input lines selected by two address line combinations $A_1A_0 = 00, 01, 10, 11$ respectively and f is the output of the multiplexer, EN is the Enable input.



The function $f(x, y, z)$ implemented by the above circuit is

- (a) xyz' (b) $xy + z$
(c) $x + y$ (d) None of these
- Q.58** A function $F(A, B, C) = \sum m(0, 1, 4, 6)$ is to be implemented using a 4 : 1 MUX with 'AC' as select lines. Which of the following gives correct input of multiplexer?

- (a) $I_0 = B, I_1 = \bar{B}, I_2 = 0, I_3 = 1$
(b) $I_0 = \bar{B}, I_1 = B, I_2 = 1, I_3 = 1$
(c) $I_0 = \bar{B}, I_1 = \bar{B}, I_2 = 1, I_3 = 0$
(d) $I_0 = B, I_1 = 1, I_2 = \bar{B}, I_3 = 0$

- Q.59** Two functions F_1 and F_2 are given as;

$$F_1 = \sum m(0, 1, 2, 3, 6, 7, 13, 15)$$

$$F_2 = \sum m(2, 3, 6, 7, 10, 11, 12)$$

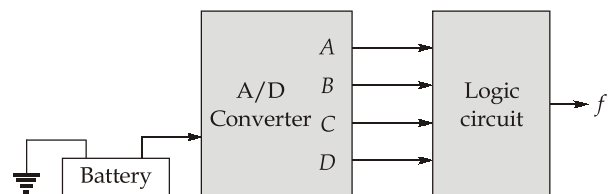
The no. of Essential Prime Implicants (EPIs) in

F_1 and F_2 are A and B . The value of $\frac{A}{B}$ is _____.

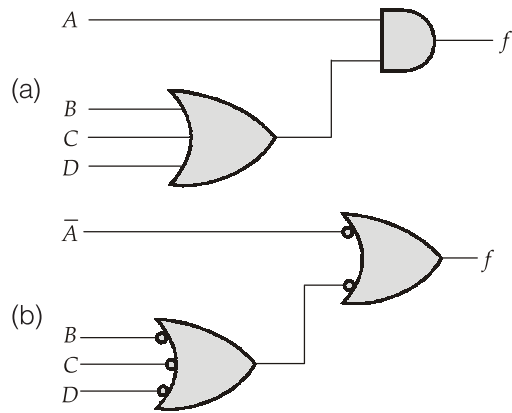
- Q.60** Total number of 2×1 multiplexers required to implement full subtractor is

- (a) five 2×1 multiplexers with available of NOT gates
(b) seven 2×1 multiplexers with available of NOT gates
(c) five 2×1 multiplexers without available of NOT gates
(d) seven 2×1 multiplexers without available of NOT gates

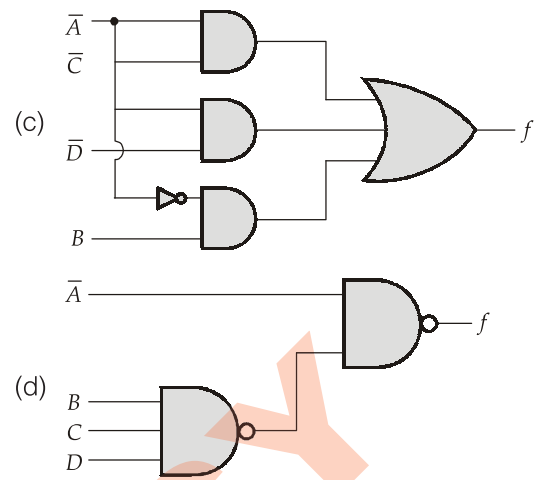
- Q.61** Consider the following analog to digital converter, monitoring the DC voltage of a +12 V storage battery. The converter's output is a four bit binary number, $ABCD$, corresponding to the battery voltage in steps of 1V, with A as the MSB. The converter's binary outputs are fed to a logic circuit that is to produce a high output as long as the binary value is greater than $(0110)_2 = (6)_{10}$; that is, the battery voltage is greater than 6 V.



Which one of the following circuits best describes the logic circuit shown in the above figure.



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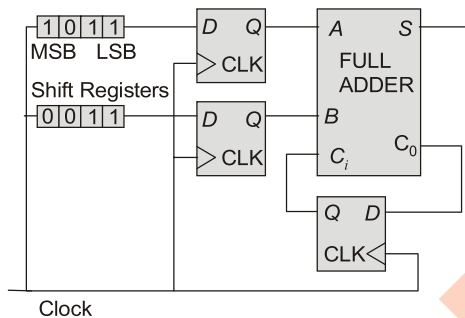
[MSQ]

■■■■

Sequential Circuits

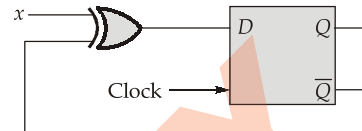
LEVEL 1 Questions

- Q.1** For the circuit shown in the figure below, two 4-bit parallel-in serial-out shift registers loaded with the data shown are used to feed the data to a full adder. Initially, all the flip-flops are in clear state. After applying two clock pulses, the outputs of the full adder should be



- (a) $S = 0$ $C_0 = 0$ (b) $S = 0$ $C_0 = 1$
 (c) $S = 1$ $C_0 = 0$ (d) $S = 1$ $C_0 = 1$

- Q.2** The circuit acts as



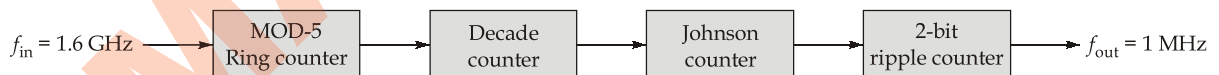
- (a) D-Flip Flop (b) T-Flip Flop
 (c) Both (a) and (b) (d) None of these

- Q.3** For the following characteristic table using XY flip-flop, the characteristic equation $Q(t+1) =$

X	Y	$Q(t+1)$
0	0	1
1	0	$Q(t)$
0	1	$\bar{Q}(t)$
1	1	0

- (a) $Y\bar{Q}(t) + \bar{X}\bar{Q}(t)$ (b) $\bar{Y}Q(t) + X\bar{Q}(t)$
 (c) $\bar{Y}Q(t) + \bar{X}\bar{Q}(t)$ (d) $YQ(t) + X\bar{Q}(t)$

- Q.4** Consider the figure shown below. The four counters are connected in cascade. The number of flip-flops used to construct the third block which is a Johnson counter is _____.



- Q.5** A MOD-8 gray up-counter is designed using T-flip-flops. If the initial state of the counter is $S_0 = 110$, then the minimum number of clock cycles required to reach a state of (101) will be _____.

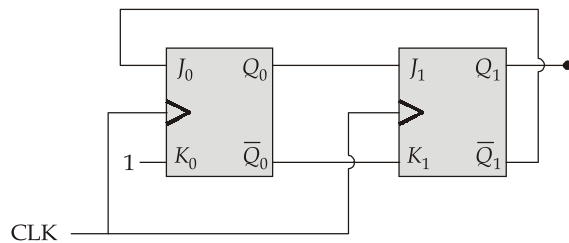
- Q.6** Consider the following statements:

- An RS Latch is a type of flip-flop with a 1-bit memory.
- An encoder has n input lines and $2n$ output lines.
- Left shift register can be termed for either division or multiplication.

Which of the above is correct statements?

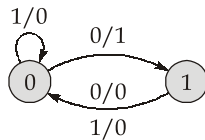
- (a) I only (b) II only (c) II and III only (d) I and III only

- Q.7** If the initial state of the counter Q_1Q_0 shown below is 01, then what is the modulus of the counter?



- (a) 1 (b) 2
(c) 3 (d) 4

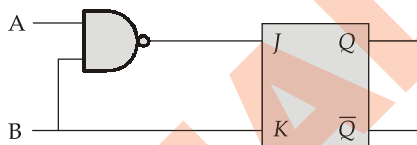
- Q.8** Consider the below state diagram:



The next state equation of the above flip-flop will be when Y represents input is

- (a) $\overline{Y - Q_n}$ (b) $Y \oplus Q_n$
(c) YQ_n (d) $\overline{Y + Q_n}$

- Q.9** An AB flip-flop is constructed from a JK flip-flop as shown in the below figure. The expression for the next state Q_{n+1} is

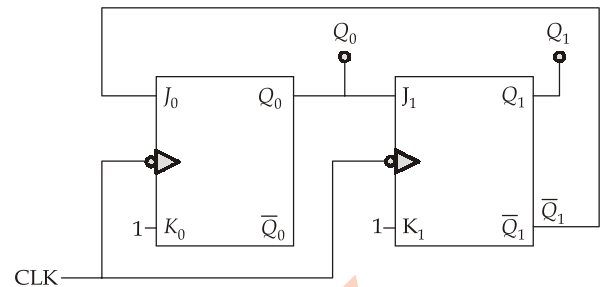


- (a) $\overline{B} + \overline{A}Q_n$ (b) $AB + A\overline{B} + \overline{A}BQ_n$
(c) $\overline{A}B + Q_n$ (d) $\overline{A}\overline{B} + \overline{A}B + \overline{A}BQ_n$

- Q.10** If the number of unused states in K-bit Johnson counter is 52, then the number of unused states when ring counter is implemented using same K-bit flip-flop is _____.

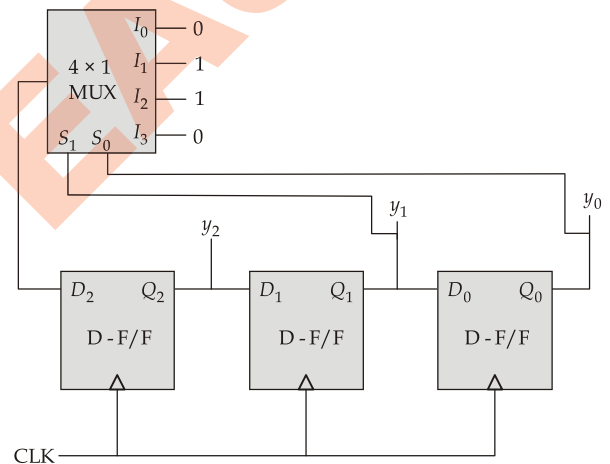
- Q.11** A modulo-8 ripple counter uses J-K flip-flops. If the propagation delay of each flip-flop is x ns and the maximum clock frequency that can be used is 4 MHz, then the value of x is _____. (Upto 2 decimal places)

- Q.12** Consider the following counter:



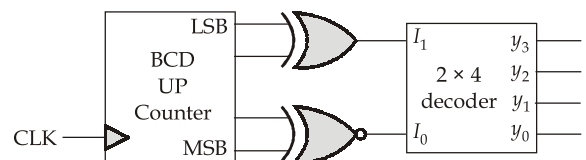
The number of states presents in above counter is _____.

- Q.13** A three bit pseudo random number generator is shown in the figure. Initially the value of output $y = y_2y_1y_0$ is set to 111.



The value of output y after three clock cycles is _____.

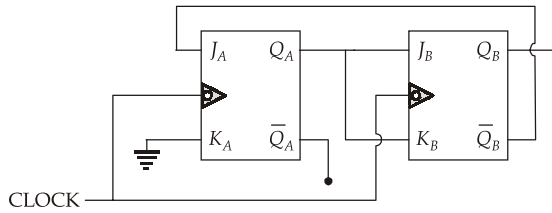
- Q.14** For the circuit shown below:



For how many number of clock pulses for which y_2 is '1'?

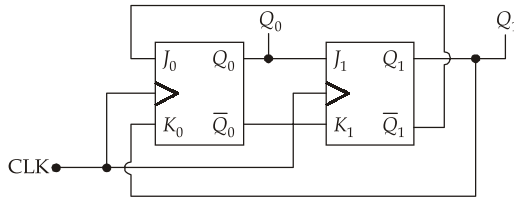
- (a) 2 clock pulses (b) 3 clock pulses
(c) 10 clock pulses (d) 6 clock pulses

Q.15 A circuit is designed with 2 – J-K FF's. If the output $\theta_A \theta_B = 10$ at starting, what will be the output ($\theta_A \theta_B$) after 13th clock pulse.



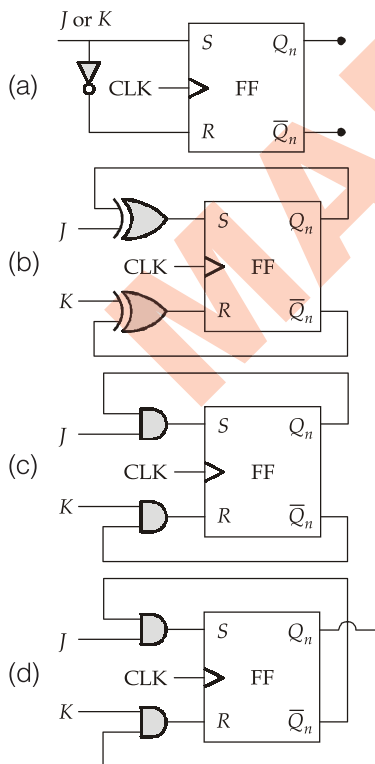
- (a) 00 (b) 10
(c) 11 (d) 01

Q.16 In the following sequential circuit, the initial state (before the first clock pulse) of the circuit is $Q_1 Q_0 = 00$. The state ($Q_1 Q_0$), immediately after the 334th clock pulse is



- (a) 00 (b) 01
(c) 10 (d) 11

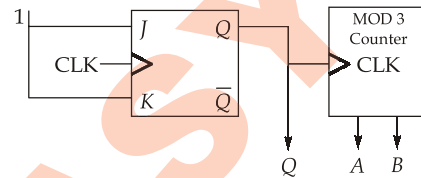
Q.17 Which of the following represents J-K flip-flop?



Q.18 The lowest frequency in kHz, if a 6 MHz clock frequency is applied to a cascaded counter of modulus-2 counter and modulus-3 counter are _____.

Q.19 For a Mod-10 counter, Johnson counter uses X FF's, ring counter uses Y FF's and ripple counter uses Z FF's. Then X + Y + Z will be _____.

Q.20 In figure, initially $Q = A = B = 0$. After five clock triggers, the states of Q, A and B will be respectively is _____.

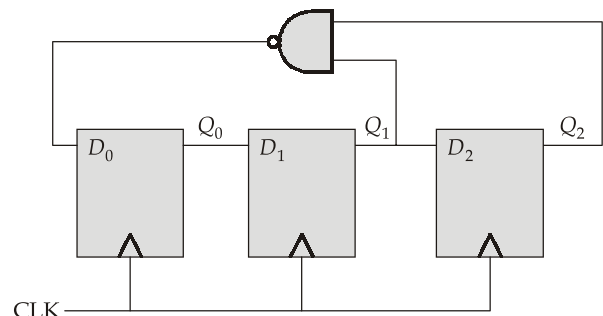


Q.21 For the following characteristic table using AB flip-flop, the characteristics equation $Q(t+1)$ is

A	B	$Q(t+1)$
0	0	$Q(t)$
0	1	$\bar{Q}(t)$
1	0	0
1	1	1

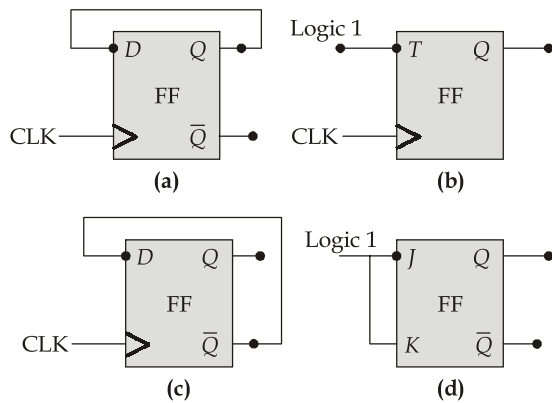
- (a) $\bar{A}\bar{B}Q(t) + B\bar{Q}(t) + AB$
(b) $\bar{A}\bar{B}Q(t) + AB$
(c) $\bar{A}\bar{B}Q(t) + \bar{A}\bar{B}$
(d) $\bar{A}\bar{B}Q(t) + \bar{B}Q(t) + AB$

Q.22 In the circuit shown below, initially all flip-flops are cleared. The output $Q_2 Q_1 Q_0$ after four clock pulses is:



- (a) 111 (b) 110
(c) 100 (d) 000

Q.23 Which of the following FF (flip-flop) is acting as a complementing flip-flop?

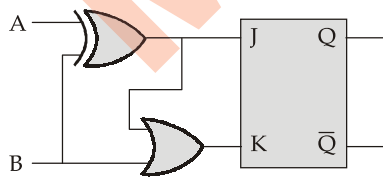


- (a) Only (a) and (b)
 (b) Only (a), (b) and (d)
 (c) Only (b), (c) and (d)
 (d) All of the above

Q.24 A 4 bit ($Q_3 Q_2 Q_1 Q_0$) binary down ripple counter is designed. The clock cycles are negative edge triggered and J-K-Flip-Flops (FF's) are used during the design process. Both the inputs J and K of FF's are connected to logic 1. What will be the input to clocks of next MSB FF's

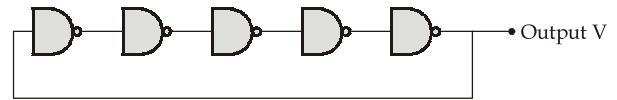
- (a) Complemented output of previous LSB FF.
 (b) Output of previous LSB FF.
 (c) Flip-Flops clock should always be 1.
 (d) None of the above

Q.25 An AB flip-flop (FF) is constructed from a JK-FF as shown in the figure. The expression for the next state Q_{n+1} is



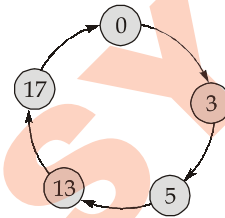
- (a) $(A \oplus B)Q_n$
 (b) $\bar{A}(B \odot Q_n) + A\bar{B}$
 (c) $\bar{A}(B \oplus Q_n) + A\bar{B}\bar{Q}_n$
 (d) $A\bar{B}Q_n + \bar{A}B$

Q.26 Each NAND gate has propagation delay of 100 nanoseconds (ns).



The frequency at output V (in MHz) is _____.

Q.27 The sequence diagram of a counter is shown in the figure below:

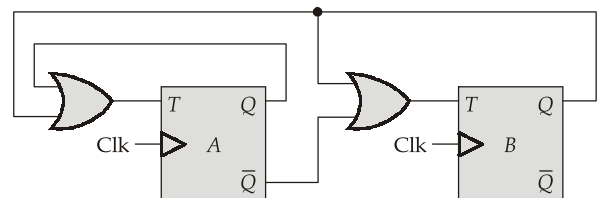


If the state value shown in the diagram are decimal equivalents. The number of FF's required to implement the counter is _____.

Q.28 The race around condition occurs in a level trigger J-K flip-flop when

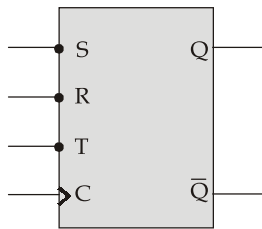
- (a) Both the inputs are 0
 (b) Both the inputs are 1
 (c) $J = 1$ and $K = 0$
 (d) $J = 0$ and $K = 1$

Q.29 The circuit shown in figure below is:



- (a) a MOD-2 counter
 (b) a MOD-3 counter
 (c) Generate sequence 00, 10, 01, 00
 (d) Generate sequence 00, 10, 00, 10, 00

Q.30 Amazon announce a new flip flop named Set-Reset-Toggle due to shortage elements to the electronics and computer industries. The device symbol and function table for this flip flop are shown below:



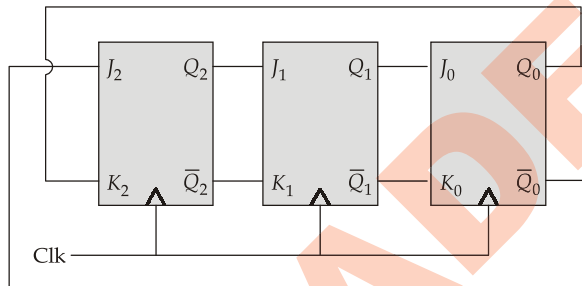
S	R	T	Function
X	X	1	Toggle
1	0	0	Set
0	1	0	Reset
0	0	0	Hold
1	1	0	Hold

Which of the following is the characteristic equation?

- (a) $Q^+ = TQ' + T'[Q(S \oplus R) + SR']$
 (b) $Q^+ = (T \oplus Q) + S + RQ$
 (c) $Q^+ = T'Q + T[Q'(S + R) + SR']$
 (d) $Q^+ = TQ' + T'[Q(S + R') + SR']$

Q.31 The minimum value of propagation delay in each Flip-Flop when a 10-bit ripple counter skip a count which is clocked at 10 MHz is _____ nsec.

Q.32 Consider the counter circuit shown in the figure below:

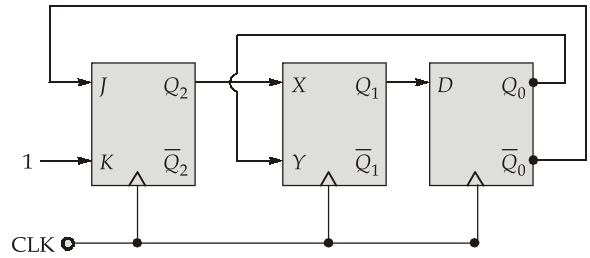


The present state ($Q_2Q_1Q_0$) of the counter before applying the clock pulse was (101). If the input clock frequency to the circuit is 100 kHz, then the output frequency of the circuit will be _____ kHz.

Q.33 A synchronous counter is designed using J-K FF, X-Y FF and D-FF as shown below. X-Y FF truth table is

X	Y	Q_{n+1}
0	0	1
0	1	$\bar{Q}_n$
1	0	0
1	1	Q_n

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If the initial content of the counter is 001 at Q_2, Q_1, Q_0 , the number of clock pulses after which the counter is back to the same state are _____.

Q.34 Consider a 4-bit Johnson counter with an initial value of 1100. The counting sequence of this counter is

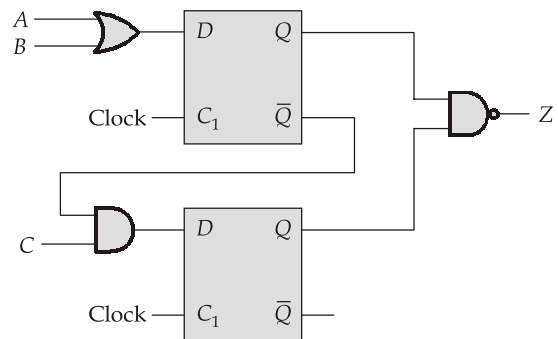
- (a) 12, 8, 0, 1, 3, 7, 15, 14, 12
 (b) 12, 11, 13, 15, 0, 1, 3, 5, 12
 (c) 12, 14, 0, 2, 4, 6, 8, 10, 12
 (d) 12, 14, 15, 7, 3, 1, 0, 8, 12

Q.35 A modulo-N ripple counter uses J-K flip-flops. If the propagation delay of each flip-flop is 2.5 nsec, then which of the statement(s) is/are correct for above given counter. (Assume the maximum clock frequency 100 MHz).

- (a) It could be decade counter
 (b) It has 3 flip-flops
 (c) It could be MOD-8 counter
 (d) It has 4-flip-flops.

[MSQ]

Q.36 Consider the following logic diagram :

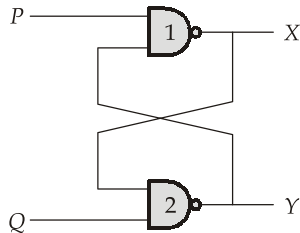


After applying 4 clock pulses, which input sequence(s) for ABC will always generate logic 1 at the output Z. (Assume initially all flip-flops are cleared)

- (a) 111
 (b) 101
 (c) 001
 (d) 110

[MSQ]

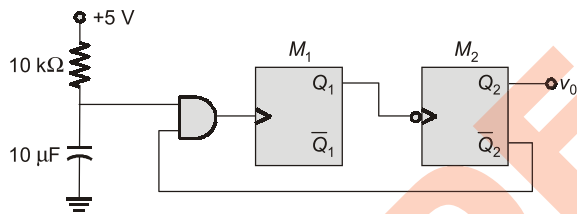
- Q.37** In the latch circuit shown, the NAND gates have non-zero, but unequal propagation delay. The present input condition is ; $P = Q = '0'$. If the input condition is changed simultaneously to $P = Q = '1'$, the outputs X and Y are



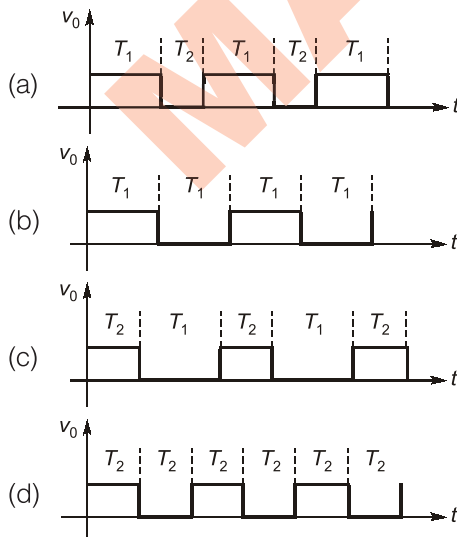
- (a) $X = 1, Y = 1$ (b) $X = 1, Y = 0$
(c) $X = 0, Y = 1$ (d) $X = 0, Y = 0$ [MSQ]

LEVEL 2 Questions

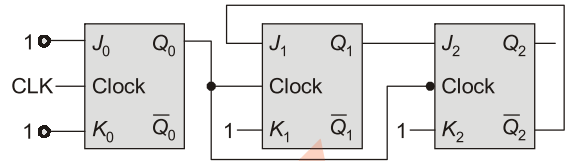
- Q.38** Two monoshot multivibrators, one positive edge triggered (M_1) and another negative edge triggered (M_2), are connected as shown in figure.



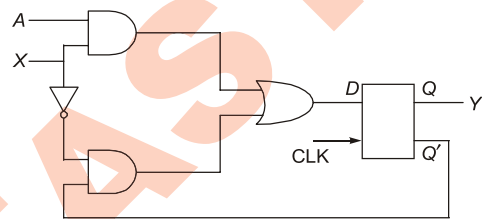
The monoshots M_1 and M_2 when triggered produce pulses of width T_1 and T_2 respectively, where $T_1 > T_2$. The steady state output voltage V_0 of the circuit is



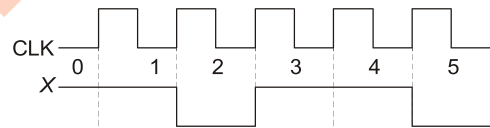
- Q.39** The figure shows a digital circuit constructed using negative edge triggered J-K flip flops. Assume a starting state of $Q_2 Q_1 Q_0 = 000$ will repeat after _____ number of cycles of the clock CLK.



- Q.40** Consider the following circuit involving a positive edge triggered D FF.

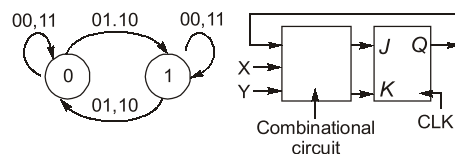


Consider the following timing diagram. Let A_i represent the logic level on the line A in the i -th clock period.



Let A' represent the complement of A. The correct output sequence on Y over the clock periods 1 through 5 is

- (a) $A_0 A_1 A_1' A_3 A_4$ (b) $A_0 A_1 A_2' A_3 A_4$
(c) $A_1 A_2 A_2' A_3 A_4$ (d) $A_1 A_2' A_3 A_4 A_5'$
- Q.41** Consider the following state diagram and its realization by a JK flip flop.



The combinational circuit generates J and K in terms of x, y and Q. The Boolean expressions for J and K are :

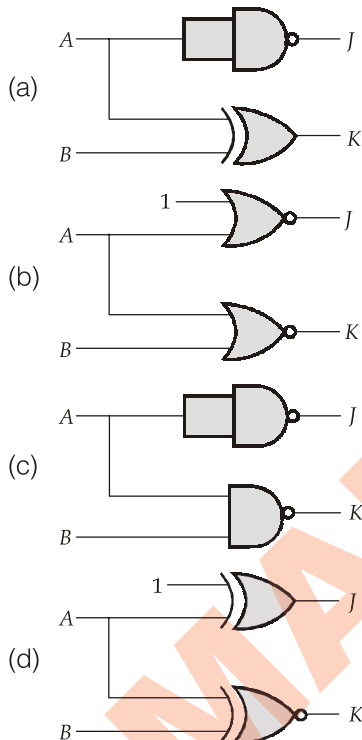
- (a) $\overline{x \oplus y}$ and $\overline{x \oplus y}$ (b) $\overline{x \oplus y}$ and $x \oplus y$
(c) $x \oplus y$ and $\overline{x \oplus y}$ (d) $x \oplus y$ and $x \oplus y$

Q.42 A new two input flip flop is designed as shown in figure. The table shows the characteristic table of the A - B flip-flop.

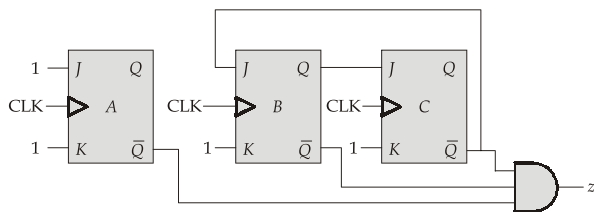
A	B	Q_{n+1}
0	0	$\overline{Q}_n$
0	1	1
1	0	Q_n
1	1	0



The combination logic is



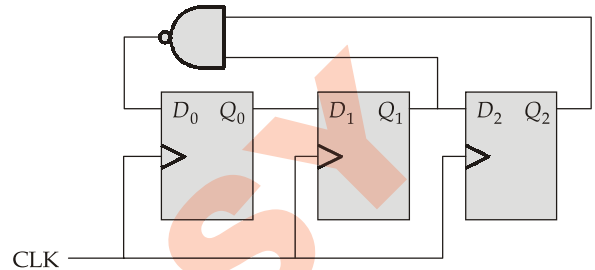
Q.43 Consider a sequential circuit using three J-K flip-flops and one AND gate shown in figure. Output (z) of the circuit becomes '1' after every N-clock cycles.



Which of the following represent the value of N?

- (a) 4 (b) 5
(c) 6 (d) 7

Q.44 In the circuit shown below, initially all flip-flops are reset. The output $Q_2Q_1Q_0$ after three clock pulses is



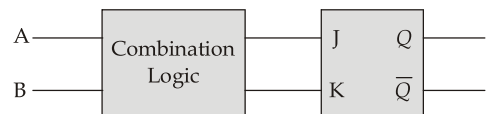
- (a) 011 (b) 111
(c) 011 (d) 110

Q.45 Consider a JK flip-flop with $J = Q'$ and $K = 1$. Assuming that the flip-flop was initially cleared and then clocked for 6 pulses. What will be the sequence of bits at the output Q?

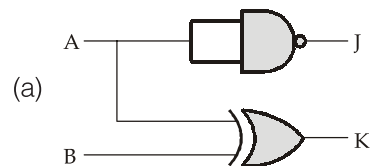
- (a) 011001 (b) 100100
(c) 101010 (d) 010010

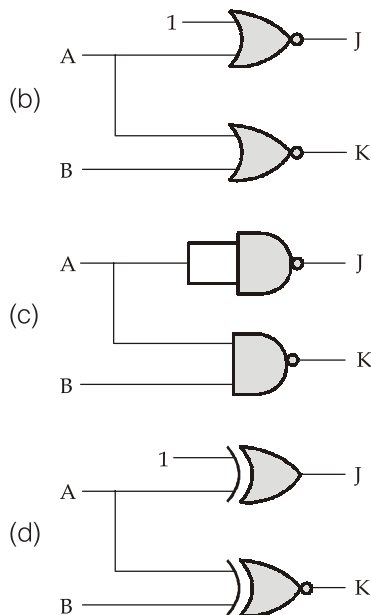
Q.46 A new two input flip flop is designed as shown in figure. The table shows the characteristic table of the A-B flip-flop.

A	B	Q_{n+1}
0	0	$\overline{Q}_n$
0	1	1
1	0	Q_n
1	1	0

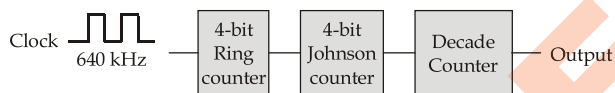


The combination logic is

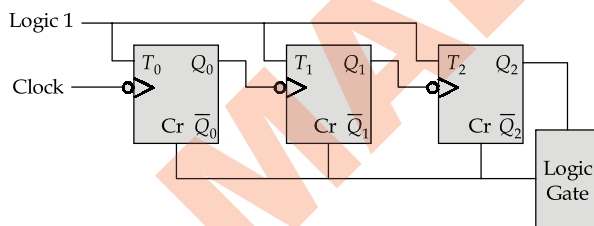




Q.47 If the input frequency applied to given system is 640 KHz then the output frequency will be _____ kHz.



Q.48 The 3-bit ripple counter (shown below) is to be designed as a MOD 4 counter

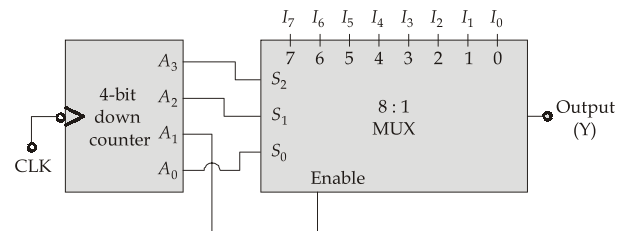


What is the best architecture of the 'Logic gate'?

- (a) a 3-bit input AND gate
- (b) a 2-input AND gate
- (c) a NOT gate
- (d) a wire connection (no logic gate needed)

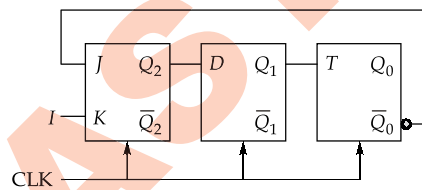
Q.49 A 4-bit Down counter is used to control the output of the multiplexer as shown in figure. The counter is initially at $(1111)_2$, then the output of the multiplexer will follow the sequence

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- (a) $I_7, 0, I_6, 0, I_5, 0 \dots$
- (b) $I_7, 0, 0, I_6, 0, 0, I_5 \dots$
- (c) $I_7, I_6, I_5, I_4, I_3 \dots$
- (d) $I_7, I_6, 0, 0, I_5, I_4, 0, 0 \dots$

Q.50 Consider the following synchronous counter made up of JK, D, T flip-flops.

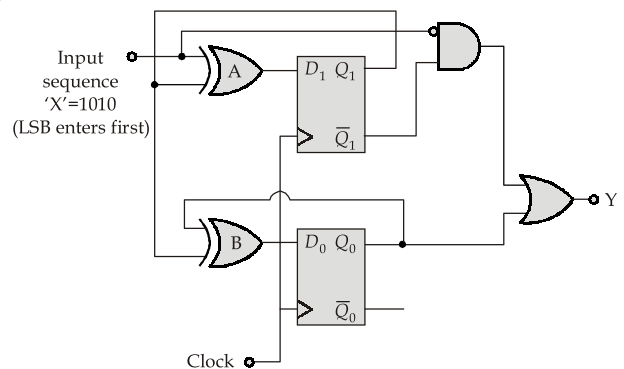


The modulus value of the counter is _____.

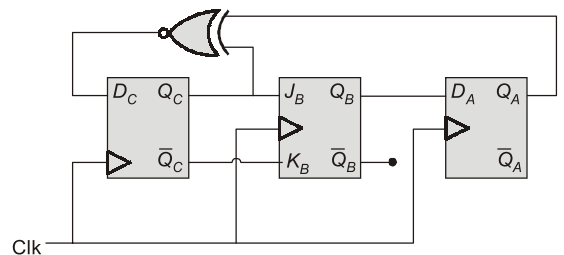
Q.51 Consider a clocked sequential circuit as shown in the figure below. Assuming initial state to be

$$Q_1 Q_0 = 00$$

For an input sequence $X = 1010$, the respective output sequence will be _____.

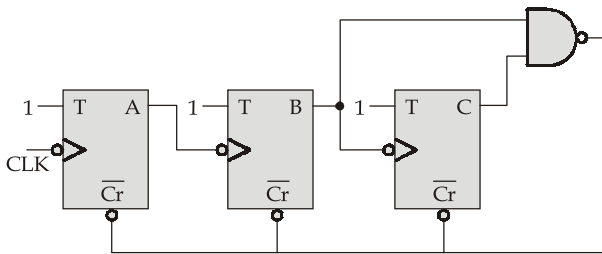


Q.52 Consider the digital circuit shown below:



If initially $Q_A Q_B Q_C$ is at 110 then minimum number of clock required to get $Q_A Q_B Q_C$ equal to 011 is _____.

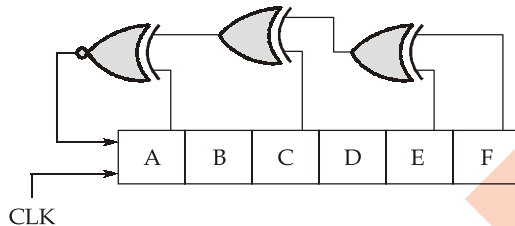
Q.53 Consider the asynchronous circuit given below:



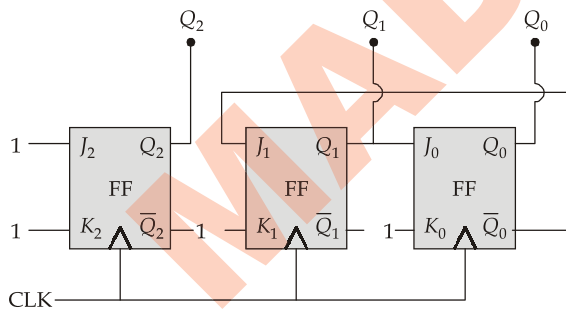
The modulus value of circuit is _____.

Q.54 Suppose a synchronous counter is designed that counts the sequence (1, 3, 1, 0, 1, 2 and repeats). The minimum number of T flip-flop are required to construct this sequential circuit are _____.

Q.55 A six bit right shift register is initialized to a value of (A, B, C, D, E, F) = (010000). The minimum number of clock pulses are required to produce 101101 are _____.

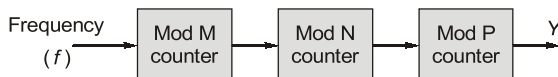


Q.56 Consider the below sequential circuit using J-K flip-flops. Initially the output at $(Q_2 Q_1 Q_0) = 001$.



The number of clock cycles required to get at $Q_2 Q_1 Q_0 = 000$ is _____.

Q.57 Consider the figure given below:



Assume frequency f is applied at M counter. All the counters M, N and P are cascaded one after another. The output is at Y after P counter. Which of the following option holds?

- (a) Mod $M + N + P$ counter, $f/(M + N + P)$ at Y.
- (b) Mod $M \times N \times P$ counter, $(f/M + f/N + f/P)$ at Y.
- (c) Mod $M \times N \times P$ counter, $f/(MNP)$ at Y.
- (d) Count is not possible because cascaded.

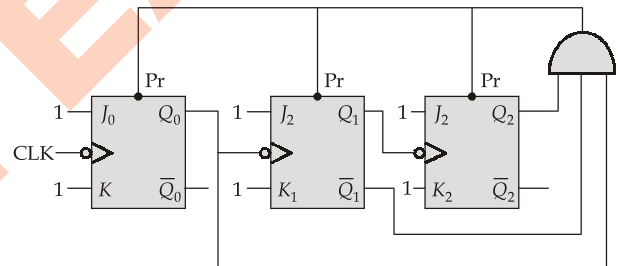
Q.58 Consider the following statements:

- (i) Number of unused states in Ring counter for N-bit is $(2^N - N)$.
- (ii) Number of unused states in Johnson counter for N-bit is $(2^N - 2N)$.
- (iii) Any count sequence can be designed with synchronous counter but not with asynchronous counter.

Which of the following statements are correct?

- (a) Only (i)
- (b) Only (ii)
- (c) Only (ii) and (iii)
- (d) All of the above

Q.59 Consider the below asynchronous sequential circuit. An active high preset button Pr is used in each J-K flip-flop. What will be the mod of the given circuit, initially all FF's are set.

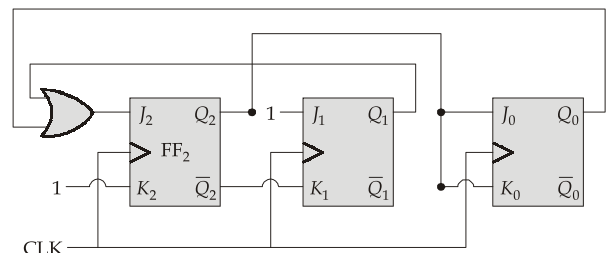


- (a) 4
- (b) 5
- (c) 6
- (d) 7

Q.60 Consider a 5 bit ring counter with an initial value of 00001. The next counting sequence of this counter is

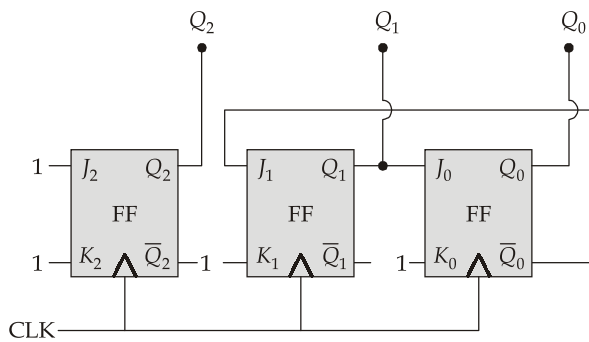
- (a) 1, 2, 4, 8, 16, 1
- (b) 1, 16, 8, 4, 2, 1
- (c) 1, 2, 3, 4, 5, 1
- (d) 1, 4, 8, 16, 2, 1

Q.61 Consider the below circuit:



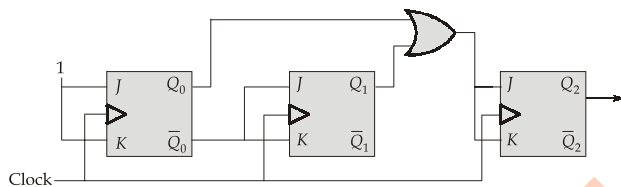
The above circuit forms a counter of MOD _____.

Q.62 Consider the below sequential circuit using J-K flip-flops. Initially the output at $(Q_2Q_1Q_0) = (001)$.

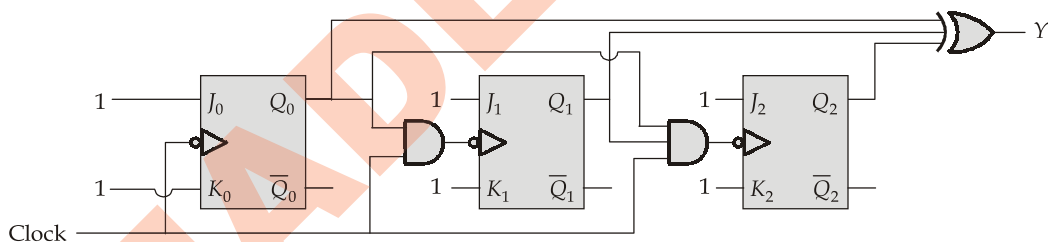


The number of clock cycles required to get at $Q_2Q_1Q_0 = 000$ is _____.

Q.63 Consider the circuit given below with initial state $Q_2Q_1Q_0 = 000$.



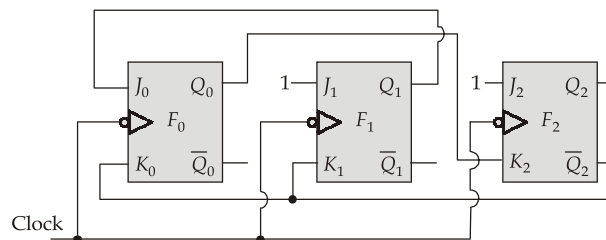
Q.65 Consider the circuit shown in the figure below:



The value of output $Q_0Q_1Q_2$ and the value of output Y after the 5th clock pulse is (Assume that initially all the flip-flop are in reset state)

- (a) Output = 001 and $Y = 1$ (b) Output = 101 and $Y = 0$
(c) Output = 100 and $Y = 1$ (d) Output = 010 and $Y = 1$

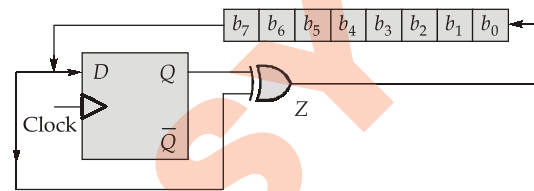
Q.66 Consider the counter circuit shown in the figure below:



Which one of the following is the correct state sequence of the circuit?

- (a) 0, 6, 3, 4, 1, 7, 2, 5, 0
(b) 0, 3, 6, 1, 4, 7, 2, 5, 0
(c) 0, 3, 6, 4, 1, 7, 2, 5, 0
(d) 0, 6, 3, 1, 4, 7, 2, 5, 0

Q.64 Consider 8-bit left shift register and D flip-flop shown in figure below is synchronized with same clock. The D flip-flop is initially cleared.



Which of the following represents the behaviour of above circuit?

- (a) Binary to 2's complement converter
- (b) Binary to 1's complement converter
- (c) Binary to Excess-3 code converter
- (d) Binary to Gray code converter

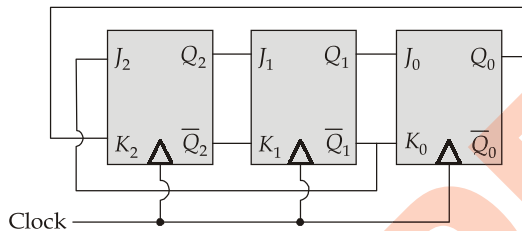
The counter is designed such that it counts the states $3 \rightarrow 4 \rightarrow 6 \rightarrow 7$, then which of

the following statements about this counter is true?

- (a) The counter enters into a lockout state if the counter starts from $(5)_{10}$
- (b) The counter enters into a lockout state if the counter starts from $(2)_{10}$
- (c) The counter enters into a lockout state if the counter starts from $(3)_{10}$
- (d) The counter do not enters into a lockout state

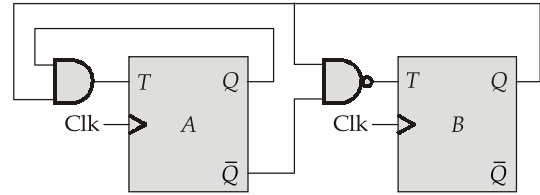
Q.67 Consider a binary counter is being pulsed by a 256 kHz clock signal. If the output frequency from the last flip-flop is 8 kHz, then the mod value of counter is _____.

Q.68 Consider the circuit shown in the figure below:



Then the value of $(Q_2 Q_1 Q_0)$ after the first clock pulse is equal to _____. (Assume all the outputs to be '0' initially)

Q.69 The circuit shown in figure below is,

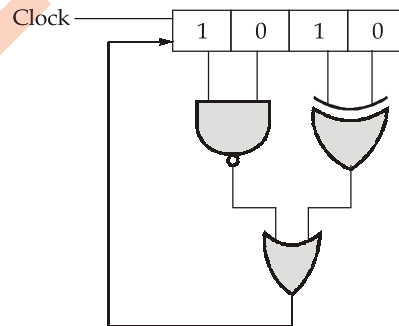


Initially both flip-flops are at 00. After two clock pulses, the content at $Q_A Q_B$ is

- (a) 00
- (b) 01
- (c) 10
- (d) 11

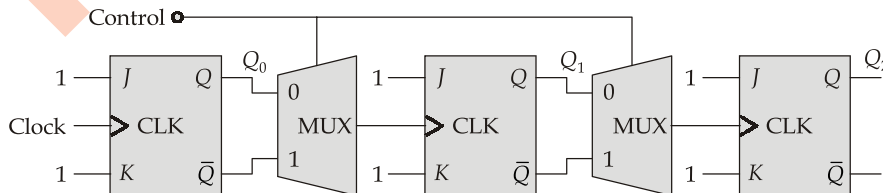
Q.70 We want to design a synchronous counter that counts the sequence 0 - 1 - 0 - 2 - 0 - 3 - 0 - 4 and then repeats. The minimum number of J-K flip-flops required to implement this counter is _____.

Q.71 A 4-bit serial-in-parallel-out right shift register with initial content as 1010 is shown in the figure below. After how many clock pulses, the content of the shift register will be 1111.



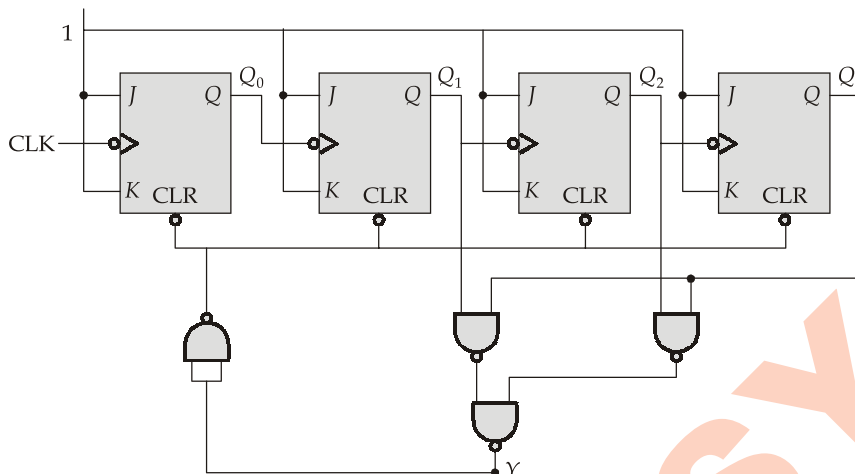
- (a) 2
- (b) 3
- (c) 4
- (d) 5

Q.72 The figure below shows a 3-bit ripple counter, with Q_2 as MSB. The flip-flops are positive edge triggered. The counting direction is



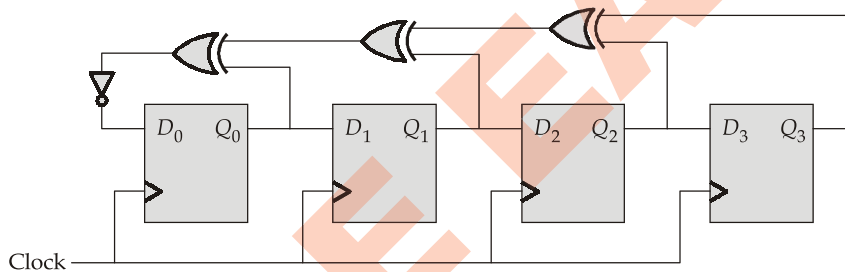
- (a) up when control = 0, down when control = 1
- (b) down when control = 0, up when control = 1
- (c) always up
- (d) always down

Q.73 Consider the counter circuit shown below,



MOD number of the given circuit is _____.

Q.74 Consider a circuit given below with initial state $Q_3 Q_2 Q_1 Q_0$ as 1010.



The number of clock pulses required to get the initial state again are

- (a) 4 (b) 5 (c) 8 (d) 10

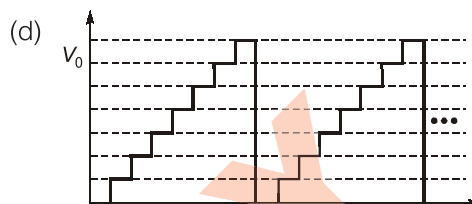
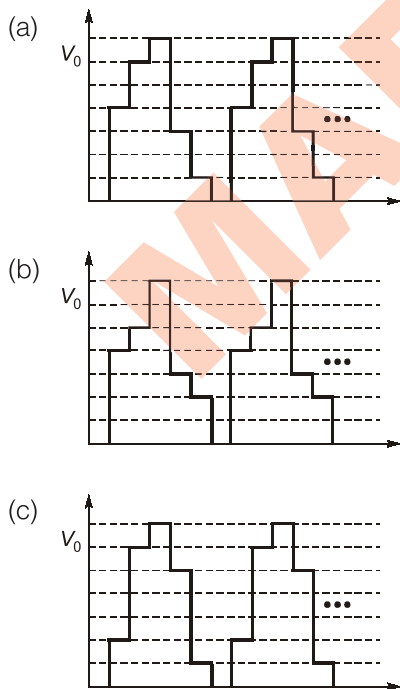
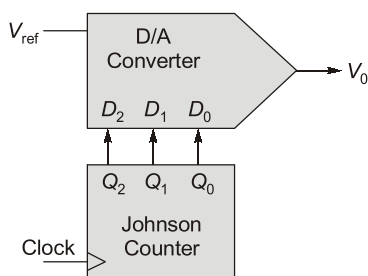
[MSQ]

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A/D and D/A Converters

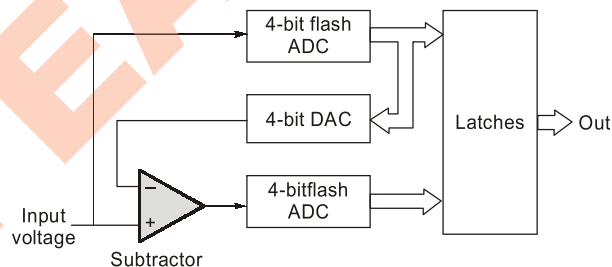
LEVEL 1 Questions

Q.1 The output of a 3-stage Johnson (twisted-ring) counter is fed to a digital-to-analog (D/A) converter as shown in the figure below. Assume all states of the counter to be unset initially. The waveform which represents the D/A converter output V_0 is



Q.2 A certain 8-bit successive approximation type analog to digital converter has full scale voltage of 2.65 V. If the conversion time $V_A = 1.5$ V is $75 \mu\text{s}$, then the conversion time for $V_A = 2$ V is ____ μs .

Q.3 Consider the circuit shown in the figure below:

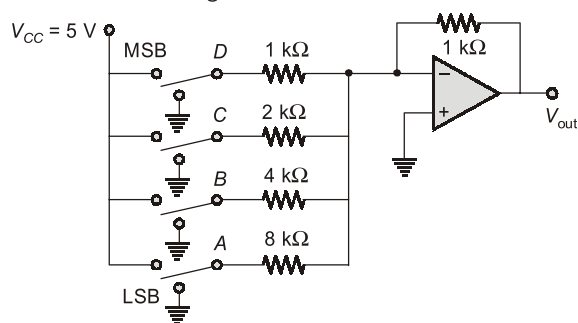


The circuit represents

- (a) 4-bit subtractor circuit
- (b) A 4-bit A to D converter
- (c) A 8-bit adder circuit
- (d) A 8-bit A to D converter

Q.4 In a 4-bit weighted resistor D/A converter, the resistor value corresponding to LSB is $64 \text{ k}\Omega$. The resistor value corresponding to MSB will be ____ $\text{k}\Omega$.

Q.5 Consider the figure shown below:



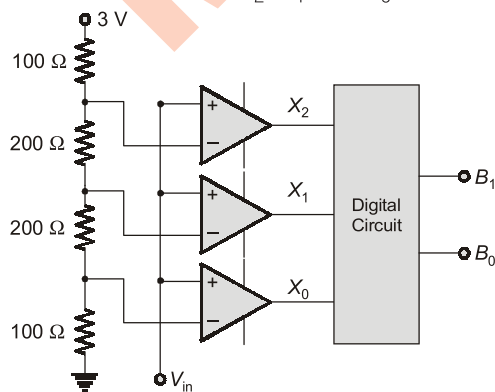
The weight of input (B) is _____ V.

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- Q.6** An 8 bit DAC produces output voltage of 2 V for a binary input of 0110 0100. The output voltage for a binary input of 1011 0011 is _____ V.
- Q.7** The ratio of maximum conversion time of 4-bit digital ramp type ADC to that of 4-bit successive approximation type ADC is ____.
- Q.8** The minimum number of bits that should be used while constructing a D/A converter to keep the quantization error below 0.4% of its full scale range is
(a) 7 bits (b) 8 bits
(c) 9 bits (d) 10 bits
- Q.9** A flash type ADC uses 15 comparators for conversion. The resolution of flash type ADC if reference voltage is 4 V will be:
(a) 0.1 V (b) 0.23 V
(c) 0.25 V (d) 0.3 V

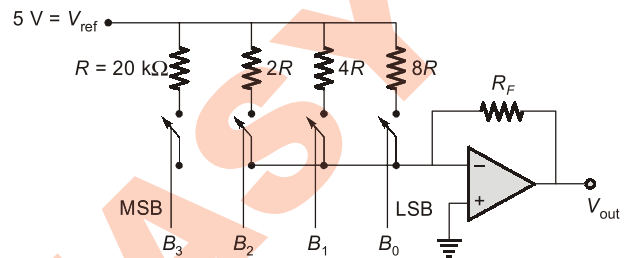
LEVEL 2 Questions

- Q.10** An 8-bit, unipolar Successive Approximation Register type ADC is used to convert 3.5 V to digital equivalent output. The reference voltage is +5 V. The output of the ADC at the end of 3rd clock pulse after the start of conversion, is
(a) 1010 0000 (b) 1000 0000
(c) 0000 0001 (d) 0000 0011
- Q.11** A 2-bit flash Analog to Digital Converter (ADC) is given below. The input is $0 < V_{IN} < 3$ volts. The expression for the LSB of the output B_0 as a Boolean function of X_2 , X_1 and X_0 is



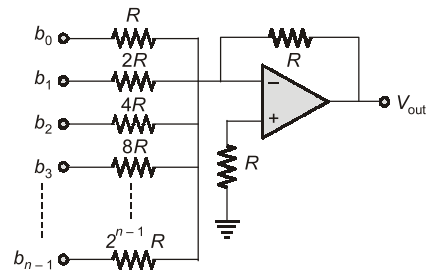
- (a) $X_0[X_2 \oplus X_1]$ (b) $\bar{X}_0[X_2 \oplus X_1]$
(c) $X_0[X_2 \oplus X_1]$ (d) $\bar{X}_0[X_2 \oplus X_1]$

- Q.12** An 8-bit successive approximation type ADC has a resolution of 30 mV. The digital output for an analog input of 2.86 V is
(a) 01011111₂ (b) 01101100₂
(c) 01100000₂ (d) 01110011₂
- Q.13** The circuit of DAC is given below. Switch are closed if input bit is 1, otherwise open.



If full scale voltage is to be $V_{out} = -4$ V, then value of R_F would be _____ k Ω . (Answer upto two decimal places).

- Q.14** A particular 6-bit DAC has a full scale output rated at 1.260 V. Its accuracy is ± 0.1 % of full scale voltage and it has an offset error of ± 1 mV. The maximum error is $\pm$ _____ mV.
- Q.15** A 3-bit resistance ladder D/A (R-2-R network) has resistor values of $R = 10$ k Ω and 20 k Ω if $V_{ref} = 8$ V, $R_f = 20$ k Ω , I_o for a digital input of 111 will be _____ mA.
- Q.16** Consider the circuit shown below, which is an n -bit DAC.



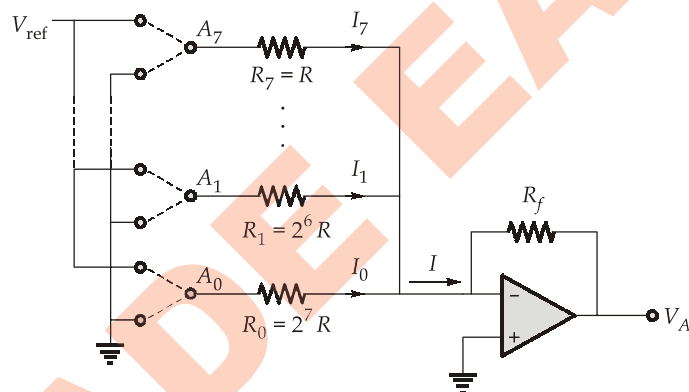
The input bits 0 and 1 are represented by 0 and 5 V respectively. If the op-amp is ideal and the maximum value of $|V_{out}|$ is 9.375 V, then the value of n will be

- (a) 4 (b) 8
(c) 12 (d) 16

- Q.17** If the input of a DAC is changed from 0111110100 to 0111111001 then output voltage change is 25 mV. Then the full scale reading of DAC is _____ V.
- Q.18** A 4-bit successive approximation type ADC has a full scale value of 15 V. The sequence of the states, the SAR will traverse, for the conversion of an input of 5.25 V is

- (a) Start Conversion → 1 0 0 0 → 0 1 0 0 → 0 1 0 1 → 0 1 0 1 → End Conversion
- (b) Start Conversion → 1 0 0 0 → 1 1 0 0 → 1 1 1 0 → 1 1 1 1 → End Conversion
- (c) Start Conversion → 1 0 0 0 → 0 1 0 0 → 0 1 1 0 → 0 1 0 1 → End Conversion
- (d) Start Conversion → 1 0 0 0 → 0 1 0 0 → 0 1 1 0 → 0 1 1 0 → End Conversion

- Q.19** Consider an 8-bit DAC using binary-weighted resistors with $V_{ref} = 5$ V.



The smallest value of R which will limit the maximum current drawn from the supply (V_{ref}) will not be more than 10 mA is

- (a) 650 Ω (b) 996 Ω (c) 1000 Ω (d) 1020 Ω

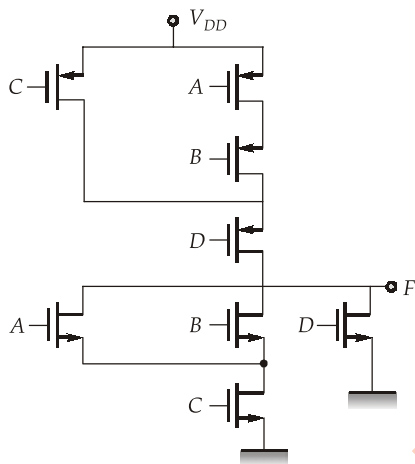
[MSQ]



Logic Families

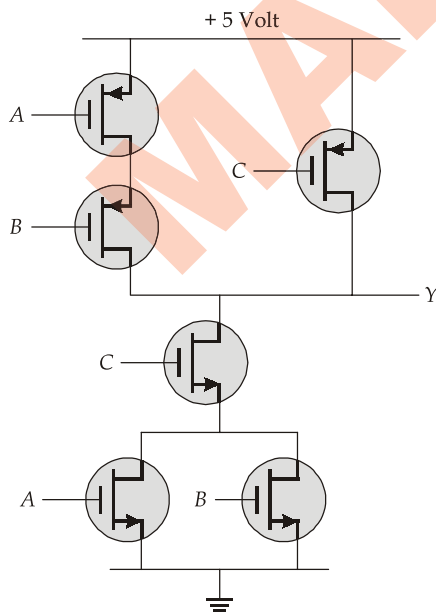
LEVEL 1 Questions

- Q.1** The output logic function (F) of the CMOS circuit shown below can be represented as

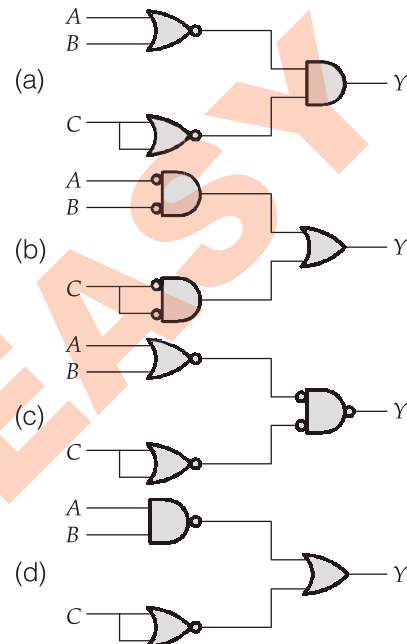


- (a) $(A + B)C + D$ (b) $\overline{(AB + C)D}$
 (c) $\overline{(A + B)C + D}$ (d) $(AB + C)D$

- Q.2** Consider the CMOS implementation



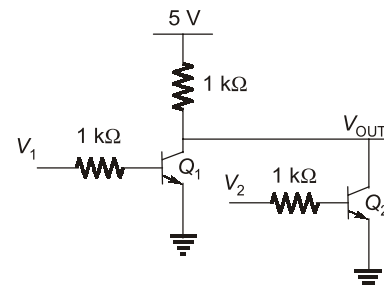
The logic circuit equivalent to the above CMOS implementation is



[MSQ]

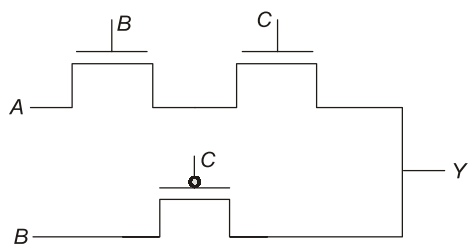
LEVEL 2 Questions

- Q.3** The logical gate implemented using the circuit shown below where, V_1 and V_2 are inputs (with 0 V as digital 0 and 5 V as digital 1) and V_{OUT} is the output, is



- (a) NOT (b) NOR
 (c) NAND (d) XOR

Q.4 The logical expression realized by the circuit shown below is



- (a) $B(A + C)$ (b) $B(A + \bar{C})$
(c) $A(B + \bar{C})$ (d) $C(A + \bar{B})$

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